M-DVP spec: detrended volume-profile engine (Keller B5)

Rev 6, 2026-11-01 (ET), approved by Ettienne Hugo 2026-11-01 (7:50 PM ET); Q17–Q21 DECIDED the same day (§14.1; Q21 = the chart start, labelled Q6 in the approval and relabelled because the rev 3 Q6 is Fib direction). Rev 6.1 (2026-11-01): PR #82 review edits (revision table). Rev 6.2 (2026-11-01): DV-13.4 slowdown budget DECIDED at ≤ 1.00 (revision table). Rev 5.3 (as on `origin/main` @ `1b8e3ba`, which adds DV-7.8) plus Ettienne's decision of 2026-11-01: the frozen per-symbol d* table is replaced by **online self-calibration** while the strategy runs (DV-3.10, `degMode = SelfCalibrate`). The decision is DECIDED; the numbers in DV-3.10 stay marked **PROBE** and are revisited only from the PR-D9a dev-window report (Q17). The approval adds the `calibStep` field (Q19). See the revision log.

Rev 5.3, 2026-10-24 (ET), approved by Ettienne Hugo 2026-10-24 (rev 5 plus the 5.1–5.3 amendments in the revision table). Rev 4 plus PR #37 review spec edits S1–S5 and the instrument-profile DV-7.0 amendment (rollMode None) (`specs/pr37-review-analyst1.md`; see the revision log). Rev 4 (2026-10-24): Rev 3 plus Ettienne's DV-3.8 decision, the DV-3.6 roll-horizon rule and removal of the 240m sweep (see the revision log). Rev 3 (2026-10-23): Rev 2 plus Ettienne's decisions on Q1, Q4, Q6 and the PS-25 primary cell. Rev 3 left nothing OPEN; rev 6 adds the OPEN items in §14.1. Rev 2 was Rev 1 plus all 16 edits (E1–E16) of Analyst 1's review (`specs/m-dvp-review-analyst1.md`), with every code claim re-checked on `origin/main` @ `629da0d`. Q2, Q3, Q5 and Q7–Q15 are resolved as Analyst 1 recommended. See the revision log below.

Implementation spec for the engine described in `specs/keller-b5-catalogue-additions.md` rev 3.1 §2 (signed off 2026-10-23) and the catalogue's §2.4b M-DVP row. Checked against `github.com/ettiennehugo/MultiCharts.Net` `origin/main` @ `629da0d` (PRs #14–#29). Every class, method and line number below was read on that commit, except where it is marked **VERIFY**. Items the book does not settle are marked **[GAP]** and are fixed here by a stated default that is listed in §14.

Sources: the repo (`_ProfileMath`, `_ProfileStats`, `_LevelRef`, `_RollDetector`, `_BarRing`, `_MtfMath`, `_PriceActionEngine`, `_Gate0`, `_LevelStudy`, `_CustomSignal`, `_VolumeProfileBase`, `_VpHistogram`, `tests/*`), `trading-idea-catalogue.md` (§2.4b L217–259; rows A2-33, PS-25, MT-07, RG-09, RG-10, KD common/KD-02/KD-06), `specs/keller-b5-catalogue-additions.md` rev 3.1 ("B5"), the Analyst 1 and Analyst 2 reviews and P6/P7 entries (superseded points are listed in §14.2), and `kindle-notes/keller-b5-dvp-analyst2.md` (Appendix E functions).

Scope. M-DVP is a **base service**, not a strategy. It fits a polynomial trend to the last `winLen` closed bars of a stream, builds a volume profile of the residuals (the "manifold"), and publishes **rails**: trend-following price levels for the next bar. It also builds a flat Fibonacci lattice from confirmed swing pivots and counts how many independent families agree at a price. It produces levels, states and journal lines. It places no orders. The consumer ideas (PS-25, MT-07, RG-09/KD-09, RG-10, KD-02, KD-06, KD-10, A2-33) are written in their own sandboxes and specs; this spec only fixes the API they read.

Not touched. The PriceAction library (`/workspace/PriceAction`) is not changed. The repo's wrapper `_PriceActionEngine` is used as it is. No `[Input]` is added to `_CustomSignal` or to any strategy. Rev 6's self-calibration settings are `DvpConfig` fields (protected config), not inputs.

Revision log

Rev	Date (ET)	Change
1	2026-10-23	First draft.
2	2026-10-23	Analyst 1 review applied: E1 score = family count (DV-6.5, DV-10.1; Q12 moot). E2 entropy call `Entropy(rows, TradedRows, 0)` (DV-4.5). E3 concentration = POC row / mean row (DV-4.5). E4 δ only on the trigger close; engine has no counting default (DV-8.2/8.3). E5 fixed row array, no `Accumulator.Dense()` (DV-0.8, DV-4.2). E6 forward coverage with the leverage term (DV-3.6). E7 verbatim 24-ratio Fib list (DV-5.3). E8 RollDetection Off ⇒ diagnostic-only (DV-7.0, DV-T27). E9 sessions/24h/half-days (DV-7.7, DV-T28/29). E10 lazy path-independence (DV-3.7/3.8, DV-T16b). E11 PS-25 primary cell PROPOSED (§10.4). E12 skip-share estimates labelled (DV-7.4). E13 `verdict41`, horizon 40 (DV-8.4, DV-11.2). E14 VA/edge/z=0/POC-tie

Rev	Date (ET)	Change
		basis (DV-4.4). E15 PR plan additions (§16). E16 conflicts l–n (§14.2). §14.1 rewritten as resolved decisions + OPEN items.
3	2026-10-23	Ettienne's decisions: Q1 = the trigger close plus 2 further closes beyond the rail; Q4 = drop 240m (MT-07 on 60m only, 240m diagnostic, no winLen 90 trial); Q6 = Fib direction pre-registered now (App E read before PR-D4 optional); PS-25 primary cell CONFIRMED (N = 2, VAH/VAL rails, EX-05 trail + 41-bar time stop). `Accepted` semantics fixed: N = further closes after the trigger when countTrigger = true; PS-25 passes countTrigger = true, N = 2; KD-06/KD-10 pass countTrigger = false (their legacy "N closes including the trigger" reading). Updated: header, DV-1.4, DV-5.3, DV-7.4, DV-7.5, DV-8.1, DV-8.3, §10.4 PS-25/MT-07/KD-06/KD-10 rows, §14.1, §14.2-i/j, DV-T19, DV-T19b, PR-D5.
4	2026-10-24	Ettienne's decision: DV-3.8 endpointDrift redefined to match PR-D3 code (previous fit evaluated at its own last bar / previous refit bar; NaN across roll or out of buffer). DV-3.6: roll-crossing forward horizons dropped, count journaled in DVP RUN. 240m winLen sweep removed (DV-1.1, DV-7.5, PR-D6).
5	2026-10-24	**APPROVED (Ettienne Hugo, 2026-10-24).** PR #37 review (Analyst 1) spec edits: S1 DV-5.1 `minBreakoutLayer = 0` in the PA config (Validate requires 0..layers−1), VERIFY resolved (XMKT `findSharedPa` pattern), host reference updated to `_CustomSignal` ~L1044 (PR #37 head). S2 DV-5.2 anchor tie rules, A selection, inclusive lookback after selection with no fallback. S3 DV-5.4 seed-based grouping (no chaining), fibNear ≤ applied after merge. S4 §2 `fibAnchor` valid = MostRecentPivot only; `ExtremeInLookback` rejected by Validate. S5 DV-0.3/§15 gen.sh extracts `// <dvp-pa>`; `_PriceActionEngine` linked whole in tests/Dvp. Instrument-profile amendment (M-INSTR rev 3 IP-9.4): DV-7.0 rollMode None → skipShare 0, stream stays tradeable; ES with RollDetection Off unchanged; different root on Data2/3 still rejected; waits on DVP host PR #36; DV-7.4/DV-11.3 consequential edits, DV-T27b/c added. Still DRAFT pending Ettienne review.
5.1	2026-10-24	Approved by Ettienne Hugo (2026-10-24), from the PR #39 review: S1 HVN/LVN match cap |Δz| ≤ 0.25 (DV-8.1); S2 ATR history depth (DV-8.1); S3 verdict41 v0 definition (DV-8.4/DV-11.2); S4 candidate slots 1:1 (DV-6.3/6.5).
5.2	2026-10-24	Approved by Ettienne Hugo (2026-10-24), from the PR #41 review: HtfAccepted signature with countTrigger/deltaAtr (§10, DV-9.4); NextRailDistAtr strictly-beyond and NaN cases (DV-9.4); Tradeable never consults diagnosticOnly (DV-9.3/DV-7.0).
5.3	2026-10-24	Approved by Ettienne Hugo (2026-10-24), from the PR #42 review: DV-12.1 halfDay on the per-session Journal line joined by day (S1); level registered after LevelStudy Bar(), first evaluated next bar (S2).
6	2026-11-01	**DECIDED (Ettienne Hugo, 2026-11-01, 5:58 PM ET):** online self-calibration replaces the frozen d* table, because a static per-symbol table does not carry over to forex, stocks or other futures. Base: `origin/main` @ `1b8e3ba` (rev 5.3 + DV-7.8). `DvpFrozenDegree`, `DevCoverageFrozen` and the offline d* freeze are removed. New DV-3.10 (scoring without look-ahead on a fixed sample grid, pending queue, rolling state of the last `calibLen` scored windows, warm-up on d = 1, monthly recheck on `tradingDayKey`, hysteresis, journal lines, determinism, holdout, cost). `degMode` = {SelfCalibrate (default), Fixed1, Calibrate}; four new `DvpConfig` fields (§2), hashed in `Canon()`. Edited: header, DV-0.1, DV-0.7, DV-1.2, DV-1.10, §2, DV-2.1, DV-2.2, DV-3.6, DV-3.7, DV-3.8, DV-8.2, DV-9.3, §10, §10.4, DV-11.1, DV-11.3, DV-13.3, DV-13.5 (new), §14.1, §14.2-o (new), DV-T8, DV-T22, DV-T23, DV-T30–T38 (new), DV-R1, DV-R3, §16 (PR-D9 rewritten, PR-D9a added, doc follow-ups). DRAFT: the DV-3.10 numbers are PROBE.
6 (approved)	2026-11-01	**APPROVED (Ettienne Hugo, 2026-11-01, 7:50 PM ET).** Q17 DECIDED: the PROBE defaults (calibLen 250, calibMinScored 60, calibMarginAbs 0.01, calibMarginRel 0.02) stand for PR-D9 and are revisited only from the PR-D9a dev-window report. Q18 DECIDED: score = the gap between pooled means (as `Calibrate` and the book). Q19 DECIDED: new protected `DvpConfig` field `calibStep` (sample step in bars, default = projBars = 60, valid ≥ 1), so 60m and daily streams can warm up. Q20 DECIDED: warming makes `Tradeable` false and a degree-change bar suppresses triggers. Q21 Chart start (labelled Q6 in the approval; relabelled Q21 in rev 6.1) DECIDED: the standard chart start for dev and holdout evaluation is 2016-01-01, because the degree path depends on the first bar. Edited: header, §2, DV-2.1, DV-2.2, DV-3.6, DV-3.10.1, DV-3.10.8, DV-13.3, DV-13.5, §14.1, PR-D9a.
6.1	2026-11-01	PR #82 review rulings (analyst, 2026-11-01): the `DVP CAL` per-year buckets are keyed by the window-end bar's trading-day year, not its calendar year (DV-3.6, new DV-T39); the chart-start decision is relabelled **Q21** (§14.1), the rev 3 Q6 Fib direction is unchanged; table fix in the rev 5.1 row (escaped |Δz|).
6.2	2026-11-01	**DECIDED (Ettienne Hugo, 2026-11-01):** "Accept the slowdown and raise the timing budget". DV-13.4 budget ≤ 0.15 (PROBE) → ≤ 1.00 (DECIDED), from the PR-D9a host runs (journal off: 6,683 ms DVP on vs 3,775 ms off over 189,478 @ES 15m bars, slowdown 0.77, DVP

Rev	Date (ET)	Change
		on Data1 + Data2, so conservative for the Data1-only budget). Edited: header, DV-13.4, DV-R3, PR-D9a.
6.3	2026-11-02	**PR-D10:** Probe P2 passed (runs R1 `8356ba60`, R2 `05c8e4d9`, R3 `c003d107`, R4 `aa78186a`, `63e2d28d`; RHM on, @ES 1m + 60m; missing-functionality-specs.md §3): `DvpConst.P2_PASSED = true`, DV-9.3 VERIFY resolved. Canon() changes (`P2_PASSED=1`), so configHash changes (golden 5f8038289e2b51f8 → 6221dc6538c1eedb); fit values unchanged.

0. Cross-cutting rules (inherited)

- **DV-0.1 Laziness.** `_DvpEngine` is an `internal sealed class` created in `StartCalc()` only when `DvpConfig.enabled` is true, and is `null` otherwise. This is the MFS §0.1 pattern used by `PriceActionConfig` / `m_pa` (`_CustomSignal` L383–397). When the engine is disabled, it costs nothing: no ring is requested, no hosted pivot engine is created, and `CalcBar` does one null check. A query while disabled returns neutral values (`null`, `NaN`, empty lists) and logs one `ETraceLevel.Warning` (`m_dvpQueryWarned`, the `m_paQueryWarned` pattern). When the engine is enabled, the fit runs **on demand**: the first query on a closed bar computes the snapshot and caches it per (stream, bar). Journal output (`journalFits`) counts as a consumer **only** when `engineSink(ETraceLevel.Journal) != null` (Q14, resolved). With the journal off and no queries, no snapshot fit runs. **Exception (rev 6):** with `degMode = SelfCalibrate` the degree scorer (DV-3.10) fits one window every `projBars` closed bars on a fixed grid, whether or not anyone queries, in the same way that `crossesRoll` and skip counting run on every closed bar (DV-7.1). Without that, the chosen degree would depend on query history. These calibration fits build no manifold, rails or lattice. Results never depend on which bars were queried (DV-3.7, DV-3.8, DV-3.10.8, DV-T16b).
- **DV-0.2 Configuration.** `protected DvpConfig DvpConfig { get; } = new DvpConfig();` on `_CustomSignal`, set in a derived strategy's ctor or `Create()`. No `[Input]`. `Validate()` returns null or an error list. `Canon()` returns every output-affecting field and constant in a fixed order, and the journal's per-run `configHash` covers it (P3 §0.5).
- **DV-0.3 MC-free core.** The fit, manifold, rails, lattice math, confluence and acceptance primitives live in `_DvpEngine.Function.CS` between `// <dvp-core>` and `// </dvp-core>`. They use no MultiCharts types, so `tests/Dvp/gen.sh` extracts them with awk (the `tests/ProfileTests/gen.sh` pattern) together with `// <profile-math>`, `// <level-ref>` and `// <dvp-pa>` (`DvpPaBridge`, MC-free; rev 5). `_PriceActionEngine.Function.CS` is linked whole into tests/Dvp (not extracted). Host wiring (rings, ATR feed, rolls, pivots, journal) lives outside the region and in `_CustomSignal`.
- **DV-0.4 C# 7.3, no per-bar allocation.** No switch expressions, `??=`, ranges, LINQ or per-bar `new` on the fit path. Work arrays (`p_{k-1}`, `p_k`, residuals, rows) are allocated once per stream in `Configure()`.
- **DV-0.5 Causality.** A rail used **during bar t** comes only from bars `t−n … t−1` (closed) and an ATR known at `t−1`. An HTF stream contributes only after its bar has closed (`HtfClosedBar`, `_MtfMath.ClosedIndex`), and only once Probe P2 has passed (DV-9.3). Pivots count from their `confirmBar` / `confirmTime`, never from `extremeBar`.
- **DV-0.6 Journal.** Facts go to `engineSink(ETraceLevel.Journal)`, stamped with bar times (`yyyy-MM-dd HH:mm:ss`, chart TZ), never bar indexes. A `null` sink formats no string.
- **DV-0.7 Recalc idempotence.** Each snapshot is a pure function of (window bars, ATR at t−1, confirmed pivots, config, degree in effect at t−1). The degree in effect is itself a pure function of the stream's closed bars from the first bar the engine is fed (DV-3.10.8). A recalculation reproduces both bit for bit.
- **DV-0.8 Reuse, not copy.** Value area uses `_ProfileMath.ValueAreaExpansion` and `PocIndex`. Rows are a fixed `double[manRows]` per stream, allocated in `Configure()` and cleared per refit. Residual rows are not tick keys, so `Accumulator` does not apply, and `Accumulator.Dense()` allocates two arrays per call (verified, `_ProfileMath` L312–322), so it is not used on the fit path. Entropy uses `_ProfileStats.Entropy`. Prominence uses a routine extracted from `Gate0Profile.Nodes` (DV-4.6). Rolls use `rollOn(ds)` / `_RollDetector` / `RingBar.rollBar` / `IsRollWindow`. Bars come from `requireRing`/`Ring`/`_BarRing`. HTF access uses M-MTF. Pivots come from `_PriceActionEngine`. ATR uses `Gate0Atr`. Levels are `LevelRef` with new `ELevelKind` values.

1. Fixed rulings (normative, from B5 rev 3.1; not configurable)

- **DV-1.1** Window `winLen = 240` closed bars by default (range 30..1000). No winLen sweep is run (240m dropped, Q4; DV-7.5).
- **DV-1.2** Degree `d ≤ 3`. It is chosen by the coverage-match score (DV-3.6) and **re-selected online** while the strategy runs (`degMode = SelfCalibrate`, DV-3.10): only from closed bars, only at the first closed bar of each calendar month, with hysteresis, and d = 1 during warm-up. Degree 1 is the sensitivity arm (`Fixed1`). BIC is computed and journaled, **report-only**. (Rev 6, DECIDED (Ettienne Hugo, 2026-11-01): replaces the frozen dev-window d*.)
- **DV-1.3** Confluence is a **weightless family count**. A qualifying cluster needs all **3** families {manifold, sigma, lattice}. Report-only weights are VA 1.00 and POC 0.77. The weights 1.00/0.80 (App F) and 1.0/0.7/0.8 (App E) are **retracted and never used**. No code path may read them (test DV-T18c).
- **DV-1.4** Acceptance: **N = 2** primary, sweep {2, 3, 5}, buffer **δ = 0.10 ATR**. For PS-25, N counts closes **after** the trigger: N = 2 means the trigger close plus 2 further closes beyond the rail (Q1 **DECIDED (Ettienne Hugo, 2026-10-23)**).
- **DV-1.5** The prediction cone is **diagnostic only**. It is never a `LevelRef`, never a confluence candidate and never a target. The band is **never faded** in a tradeable idea. KD-02 is a G0-b canary only.
- **DV-1.6** Far nodes (|z_node| ≥ 2) are used only **after acceptance** (KD-10).
- **DV-1.7** A fit window that crosses a roll is **skipped**. A stream whose skip share is above **30%** is **diagnostic-only** (DV-7.4). There is no back-adjustment.
- **DV-1.8** The ES prior is 0 (no prior edge is assumed for any DVP consumer).
- **DV-1.9** Holm family m = 5 {PS-01, PS-02, DV-02, A2-31, PS-25}, α = 0.05 (the consumers' study rule; the engine only journals what they need).
- **DV-1.10** Constants (in `Canon()`, not fields): `confMode = FamilyCount`, `minFamilies = 3`, `reportWeights = {VA 1.00, POC 0.77}`, `rowFloor = 1 tick`, `rollAdjust = None`, `minBars = 12`, `ATR_LEN = 20` (Wilder, `Gate0Atr`; Q7 resolved), `HOLD_BARS = 40` (MFE/MAE horizon). Rev 6 adds `CALIB_K = 2` (band multiple for cov_in and cov_fwd, DV-3.6) and `CALIB_RECHECK = calendar month of tradingDayKey` (DV-3.10.5). `HOLDOUT_FROM = 2025-01-01` applies to `Calibrate` only (DV-3.10.9).

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2. Configuration (`DvpConfig`, catalogue §2.2 defaults)

Field	Default	Valid	Meaning
`enabled`	false		engine created only when true
`streams`	{1}	each ≥1, `requireDataStream`	one fit per stream (Data1 LTF; Data2/3 HTF, DV-9)
`src`	HL2	HL2, Close	fitted series
`winLen`	240	30..1000	n
`degMode`	SelfCalibrate	SelfCalibrate, Fixed1, Calibrate	SelfCalibrate = online degree selection (DV-3.10); Fixed1 = sensitivity arm, d = 1; Calibrate = fit d = 1..dCap side by side, journal coverage, report d = 1 (DV-3.6)
`degMax`	3	1..3; ≥ 2 with SelfCalibrate	
`calibLen`	250	20..2000	SelfCalibrate: rolling state = the last calibLen scored windows per (stream, winLen) (DV-3.10.3). **PROBE**
`calibMinScored`	60	1..calibLen	SelfCalibrate: scored windows needed before warm-up can end (DV-3.10.4). **PROBE**
`calibMarginAbs`	0.01	0..1	SelfCalibrate: absolute score margin for a switch (DV-3.10.6). **PROBE**
`calibMarginRel`	0.02	0..1	SelfCalibrate: relative score margin for a switch, as a fraction of the current d's score (DV-3.10.6). **PROBE**
`calibStep`	60 (= the projBars default)	1..500	Sample grid step in bars for SelfCalibrate and Calibrate (DV-3.10.1). Smaller than projBars gives overlapping horizons, so slow streams (60m, daily) warm up sooner (Q19 DECIDED (Ettienne Hugo, 2026-11-01)). A field cannot default to another field, so the default is the

Field	Default	Valid	Meaning
			number 60; a config that changes projBars and wants the old grid sets calibStep to the same value.
`weightVol`	false		true = volume weights $w_i = v_i$; default $w_i = 1$
`minR2`	0.35	0..1	**flag only** (`lowR2` in journal), never a filter
`manRows`	64	8..256	rows over ±manSpan σ
`manSpan`	3	1..6	in σ
`vaPct`	70	50..95	passed to `ValueAreaExpansion` as **0.70** (fraction, see DV-4.4)
`nodeProm`	0.12	0.01..0.5	prominence as a fraction of the tallest row
`maxNodes`	3	0..10	per side of POC, per kind
`sigmaList`	{1,2,3}	>0	sigma rails at ±k
`projBars`	60	1..500	cone horizon (diagnostic)
`coneMode`	Prediction	Prediction, Confidence	√(1+h) vs √h
`coneMult`	2	>0	
`useFib`	true		lattice on/off; off ⇒ no qualifying confluence (all 3 families needed)
`fibRanges`	3	1..4	strengths fibPivot·2^(j−1) = 20/40/80; 4 allowed but **not swept** (Q10)
`fibPivot`	20	≥2	
`fibLookMult`	8	≥1	anchor lookback = fibLookMult × strength bars
`fibAnchor`	MostRecentPivot	MostRecentPivot	only MostRecentPivot is defined (DV-5.2). `ExtremeInLookback` is undefined; Validate rejects it (rev 5).
`fibMerge`	0.40	ATR	merge distance within a range
`fibNear`	14	ATR	keep levels within ±fibNear ATR of C_{t−1}
`confTol`	0.30	ATR	cluster tolerance
`splitFlow`	false		up/down-tick manifolds (`RingBar.up/down`); diagnostic
`refitEvery`	1	≥1	refit cadence in closed bars (between refits the frozen fit is re-evaluated at the new u)
`journalFits`	true		DVP FIT lines (DV-11.1)
`journalRails`	true		DVP RAIL lines (DV-11.2)

- **DV-2.1** `Validate()` rejects out-of-range values, duplicate streams, `winLen < minBars`, and `streams` entries that do not exist (`requireDataStream`). It throws `ArgumentException` in `StartCalc`, as `startExtraPa` does. Rev 6 adds: `calibLen` outside 20..2000, `calibMinScored` outside 1..calibLen, a margin outside 0..1 or NaN, `calibStep` outside 1..500 (Q19), and `degMode = SelfCalibrate` with `degMax = 1` ("DVP: SelfCalibrate needs degMax ≥ 2; use Fixed1"). The calib fields are validated in every mode, so a bad value never hides behind `Fixed1`.
- **DV-2.2 Degree selection (rev 6).** There is **no per-symbol degree table**. `DvpFrozenDegree` and `DevCoverageFrozen` are removed (DECIDED (Ettienne Hugo, 2026-11-01): a static table does not carry over to forex, stocks or other futures). The degree is chosen online per (stream, winLen) by DV-3.10. `Canon()` hashes the selection **rule**, not a result: `degMode`, `degMax`, `projBars`, the four `calib*` fields and `calibStep` (Q19), `CALIB_K` and `CALIB_RECHECK`. Two runs with the same `Canon()` and the same bars from the same first bar give the same degree path (DV-3.10.8). The rev 3 Q3 resolution (constant table hashed in `Canon()`) is superseded (§14.1).

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3. Fit (`DvpFit`, book App E `buildTrend` / `evalTrend` / `evalLeverage` / `trendSE`)

Notation: the window is bars i = 0..n−1, where i = n−1 is bar t−1. y_i = src(bar i) (HL2 = (H+L)/2). w_i = 1, or v_i if `weightVol`.

- **DV-3.1 Coordinates.** ctr = (n−1)/2, scl = (n−1)/2, u_i = (i − ctr)/scl ∈ [−1, 1]. Rails for bar t are evaluated at **u_t = 1 + 2/(n−1)** (one bar beyond the window).
- **DV-3.2 De-mean.** ȳ = Σw y / Σw; fit ỹ_i = y_i − ȳ. This gives shift invariance (test DV-T2).
- **DV-3.3 Forsythe recurrence.** $p_{-1} = 0$, $p_0 = 1$, p_{k+1}(u) = (u − A_k) p_k(u) − B_k p_{k−1}(u), where N_k = Σ w p_k^2, A_k = Σ w u p_k^2 / N_k, B_0 = 0, B_k = N_k / N_{k−1}, C_k = Σ w ỹ p_k / N_k, for k = 0..d. ŷ(u) = ȳ + Σ_k C_k p_k(u). The coefficients (ȳ, ctr, scl, A_k, B_k, N_k, C_k) fully define the fit and are journaled (DV-11.1).
- **DV-3.4 SSE and σ.** SSE_d = Σ w $ỹ^2$ − Σ_{k≤d} C_k^2 N_k. Assert SSE_d ≥ −ε·ΣwỸ² (ε = 1e−12). If it is below that, or the recurrence breaks down (N_k ≤ 0), recompute SSE by a second pass over residuals and journal `DVP SSE-FALLBACK` (Debug). σ = √(SSE_d / (n − d − 1)), using Σw in place of n when `weightVol` (**VERIFY** against App E). R^2 = 1 − SSE_d / ΣwỸ².
- **DV-3.5 Leverage, cone, BIC.** h(u) = Σ_{k≤d} $p_k(u)^2$ / N_k. The cone (diagnostic) is ŷ(u) ± coneMult·σ·√(1 + h(u)) for Prediction, or √h(u) for Confidence, for u up to 1 + 2·projBars/(n−1). BIC_d = n·ln(SSE_d/n) + (d+2)·ln n, computed for every d ≤ dCap (one pass gives all of them, since the recurrence is nested). It is journaled as `bic[]` and never used for selection.
- **DV-3.6 Degree.** dCap = min(degMax, max(1, ⌊n/8⌋)); if n < minBars there is no fit. `Fixed1` ⇒ d = 1. `SelfCalibrate` (default) ⇒ d = the degree in effect for the stream (DV-3.10), always ≤ dCap. `Calibrate` ⇒ every d ≤ dCap is fitted and journaled, and the reported d is 1 (a study mode). **Coverage-match score** (shared by `Calibrate` and `SelfCalibrate`): windows are sampled on the fixed grid of DV-3.10.1 (every `calibStep` bars, default 60 = projBars; Q19). For each sampled window and each d ≤ dCap, cov_in(d) is the share of window bars with |y − ŷ_d| ≤ kσ_d, and cov_fwd(d) is the share of the next H closed bars with |y_{n+h} − ŷ_d(u)| ≤ kσ_d√(1 + h_d(u)), u = 1 + 2h/(n−1), h = 1..H. This is the book's forward band with k = `CALIB_K` = 2 and H = projBars = 60 (B5 ch 2.2; the printed 28.2%/27.7% use this definition). Over a set of scored windows, score(d) = |mean cov_fwd(d) − mean cov_in(d)|: the absolute gap between the pooled means, as `DvpCalib.Bucket.Score` computes it on main, not a mean of per-window gaps (a 60-bar cov_fwd is too noisy to score alone). A window whose forward horizon crosses a roll splice (any bar s+1..s+H with `RingBar.rollBar`) is dropped whole, and the count is journaled in the `DVP RUN` line (rev 4; `rollDropped`). The flat-band variant (no leverage) is journaled next to it, report-only (Q2 resolved). In `Calibrate`, the set is every scored window of the run, pooled per trading-day year (the year of the window-end bar's `tradingDayKey`, so a Globex window ending on the evening of Dec 31 counts in the next year; ruling on the PR #82 review, 2026-11-01) and over all years in the `DVP CAL` lines, and windows ending on or after `HOLDOUT_FROM` are not sampled (KD-08's diagnostic run reads these lines; DV-PR8). In `SelfCalibrate`, the set is the last `calibLen` scored windows, and d is chosen by argmin score with ties to the lower d, under the hysteresis and cadence rules of DV-3.10. (Rev 6: the offline d* freeze from a dev-window study is removed.)
- **DV-3.7 Refit cadence.** With `refitEvery = r > 1`, the bars between refits re-evaluate the frozen coefficients at u = (i_t − ctr)/scl, measured from the last refit window. endpointDrift is reported only on refit bars. Refit bars are defined by bar index (`CurrentBar % refitEvery == 0` on the stream), never by when a query arrives. A query between refit bars recomputes the last scheduled refit from the ring if it is not cached. A recomputed fit uses the degree that was in effect at its own last bar (DV-3.10.7), not the current one. The self-calibration grid (DV-3.10.1) does not depend on `refitEvery`.
- **DV-3.8 Repaint mitigation.** Every refit journals the as-of fit (DV-11.1) and `endpointDrift` = (ŷ_now(u of b) − ŷ_prev(u of b)) / ATR_{t−1}, where b is the **last bar of the previous fit's window** (the previous refit bar; with refitEvery = 1 that is bar t−1 relative to the new fit's last bar t, evaluated at u = 1 in the previous fit). It is NaN across a roll splice or when b has left the ring buffer. (Rev 4, DECIDED (Ettienne Hugo, 2026-10-24): matches the PR-D3 code and test TDriftR4; replaces the earlier "u_t of the previous fit" wording.) endpointDrift is reported only on refit bars. A **degree-1 shadow fit** (d = 1, same window) is always computed alongside, because RG-10 reads its slope and KD-08 its drift. Its marginal cost is zero, since it falls out of the nested recurrence. If the previous fit is not cached, `endpointDrift` recomputes it from the ring, so the value does not depend on journal or query history (test DV-T16b). **Rev 6, degree changes:** endpointDrift always

evaluates the previous fit at the **current** fit's degree (the fit is nested, so the previous window's coefficients for every d ≤ dCap are available). It therefore measures data repaint, not the model change at a monthly switch. The first refit after a switch carries `degChanged=1` on `DVP FIT`.

- **DV-3.9 Fit diagnostics** (journal-only): r2, Durbin–Watson dw on residuals, skew of residuals, sigmaAtr = σ / ATR_{t−1}, slopeEnd = dŷ/di at i = n−1 = (dŷ/du)·(2/(n−1)), and `lowR2` = r2 < minR2.
- **DV-3.10 Online self-calibration (`degMode = SelfCalibrate`, rev 6).** DECIDED (Ettienne Hugo, 2026-11-01): the degree is calibrated while the strategy runs, from closed bars only, instead of from a frozen per-symbol table. One calibrator per (stream, winLen) and per engine config (so the RTH arm of DV-7.6 and each HTF stream of DV-9 have their own). It lives in `// <dvp-core>` (MC-free) and is fed by the host on every closed bar of its stream.
- **DV-3.10.1 Sample grid and pending queue.** Let pos be the 0-based index of the closed bar on the stream, counted from the first bar the engine is fed. A window is sampled at every pos with pos mod `calibStep` = 0 (default 60 = projBars; Q19 DECIDED (Ettienne Hugo, 2026-11-01)) whose full window (n = winLen closed bars ending at pos) is in the ring and does not cross a roll (DV-7.1). A grid position that fails either test is not sampled (`gridSkipped`), and the next candidate is pos + calibStep. The grid is fixed: it does not move when a sample is skipped, and it does not depend on queries, the journal or `refitEvery`. At a sampled pos the calibrator fits the window once to dCap (one nested recurrence gives every d), stores (ȳ, ctr, scl, A_k, B_k, N_k, C_k, σ_d) for d = 1..dCap and computes cov_in(d), and puts the entry in a **pending queue**. If a cached snapshot fit already exists for that window, it is reused. Horizons are H = projBars. With the default calibStep = projBars, samples are H apart, so at most 2 entries are pending at once (one completes on the bar the next is sampled). With a smaller calibStep the horizons overlap and at most ⌊H/calibStep⌋ + 2 entries are pending; the queue is a fixed array of that size allocated in `Configure()`. The horizon stays H = projBars for every calibStep.
- **DV-3.10.2 Scoring without look-ahead.** On each closed bar t, every pending entry adds bar t to its forward counts for each d (the band of DV-3.6). An entry sampled at s is **scored only on bar s + H**, once all H forward bars have closed, and then leaves the queue. If any bar s+1..s+H is a roll bar, the entry is dropped whole (`rollDropped`). A bar with a NaN or non-finite source value drops every pending entry (`badDropped`). A score therefore uses only bars ≤ t, and the degree in effect at bar t uses only windows scored on bars ≤ t (test DV-T30).
- **DV-3.10.3 Rolling state.** Per calibrator, a fixed-count ring of the last `calibLen` = 250 scored windows (**PROBE**), holding cov_in(d) and cov_fwd(d) for d = 1..dCap (250 × 3 × 2 doubles ≈ 12 KB). A fixed count is used, not an EWMA, because it is exact and easy to reproduce offline: the D9a check can recompute it from the `DVP CAL` inputs, and the result does not depend on a decay constant or a start-up bias. At each recheck the means are summed afresh over the ring in a fixed order, so there is no running-sum drift. score(d) = |mean cov_fwd(d) − mean cov_in(d)| over the ring (DV-3.6). n = the number of windows in the ring (≤ calibLen). For scale (arithmetic estimate, **PROBE**): @ES 15m ETH has about 92 bars per session, so about 1.5 windows per session and 32 per month; 250 windows is about 8 months. On 60m it is about 8 windows per month (250 ≈ 2.6 years). On 15m RTH stocks it is about 9 per month.
- **DV-3.10.4 Warm-up.** Until the first recheck at which n ≥ `calibMinScored` = 60 (**PROBE**; about 2 months of windows on ES 15m, about 7–8 months on 60m), the stream uses **d = 1** and is **warming**: `Snapshot.warming = true`, and `Tradeable(ds)` returns false (DV-9.3). Warm-up is known at bar time, unlike the run-level `diagnosticOnly`. One `ETraceLevel.Warning` is journaled per calibrator on its first bar: `DVP DEGREE warming stream(Data{ds}) winLen({n}) minScored({calibMinScored}): d = 1, not tradeable until a monthly recheck with n >= {calibMinScored}`. On ES 15m warm-up ends after about 2–3 months (winLen + H bars before the first score, then 60 scored windows, then the next month start).
- **DV-3.10.5 Recheck cadence.** The degree is re-selected **only** at the first closed bar of each calendar month of `tradingDayKey` on the stream's own bars: the first closed bar whose `tradingDayKey(t)` year-month differs from the previous closed bar's. On the futures base this is the Globex date (DV-7.7), so the Sunday-evening session that opens a new month counts in the new month. The stream's first bar is never a recheck. Between rechecks d never changes, so the band shape is stable within a month. Order on a closed bar t: (1) pending entries take bar t and any that reach H are scored into the ring (DV-3.10.2); (2) if t is a recheck bar, select (DV-3.10.6); (3) if t is a grid position, sample (DV-3.10.1). The fit for the window ending at t (the snapshot used during bar t+1) uses the degree after step (2).

- **DV-3.10.6 Selection and hysteresis.** At a recheck with n ≥ calibMinScored: b = argmin over d = 1..dCap of score(d), scanning upward with a strict `<`, so exact ties go to the lower d. Let c be the current degree (1 when leaving warm-up). Switch c → b only if b ≠ c and score(c) − score(b) ≥ max(`calibMarginAbs`, `calibMarginRel` × score(c)) (defaults 0.01 and 0.02, **PROBE**: about 1–2 standard errors of a paired score gap over 250 windows; to be checked in D9a). Otherwise keep c. So a lower d that only ties the incumbent does not replace it; the tie rule decides the candidate, and the margin decides the switch. Warm-up ends at the first recheck with n ≥ calibMinScored, whether or not d changes. At a recheck with n < calibMinScored, d stays 1.
- **DV-3.10.7 Journal.** Every recheck writes one line to `engineSink(ETraceLevel.Journal)`: `DVP DEGREE sel stream(Data{ds}) winLen({n}) from({c}) to({d}) scores[{s1},{s2},{s3}] n({n})`, with scores for d = 1..dCap in invariant culture `R` format (`NaN` when the ring is empty). It is stamped with the recheck bar's time like every journal line (DV-0.6). A change also writes `DVP DEGREE change stream(Data{ds}) winLen({n}) from({c}) to({d}) n({n})` at `ETraceLevel.Info`. As for all journal output, each line is one `File.AppendAllText` call, and a `null` sink formats no string. For lazy recomputes (DV-3.7, DV-3.8, DV-8.1), the calibrator keeps a degree history: a byte per ring slot, d in effect after bar pos, written at index pos mod ring capacity. A fit of a window ending at bar b uses the history entry for b.
- **DV-3.10.8 Determinism and path independence.** The degree path is a pure function of the stream's closed bars from the first bar the engine is fed (the chart's first bar after MaxBarsBack), and of `Canon()`. A recalculation reproduces it bit for bit. **Implication:** results depend on the chart's start date (it sets the grid phase and what the ring has seen), so a backtest and a live chart give the same degree path, and so the same rails, only when they start on the same first bar with the same data. A chart that starts later can choose different degrees for months. Data later replaced by a vendor backfill can also change the path. **Standard chart start (DECIDED (Ettienne Hugo, 2026-11-01)): 2016-01-01** for every dev and holdout evaluation chart. **E10 / lazy cache:** the calibrator runs on its grid regardless of queries (DV-0.1 exception), and lazily recomputed snapshots read the degree history, so journal-on, journal-off and random-query runs give identical degree paths, snapshots, endpointDrift and skipShare (DV-T16b, DV-T36). **endpointDrift:** evaluated at the current degree (DV-3.8), so a monthly switch does not show up as drift.
- **DV-3.10.9 Holdout.** The rule that Jan 2025–Oct 2026 is never read for calibration was there to stop a d* chosen with sight of the holdout. Online calibration uses only bars before each point, so at any bar in the holdout the degree comes from earlier bars only, exactly as it would in live trading; it is part of the strategy being tested, like any adaptive indicator. `SelfCalibrate` therefore reads every closed bar and ignores `HOLDOUT_FROM`. `Calibrate` (the offline study) keeps it. **The dev/holdout split still governs evaluating strategies built on DVP:** consumers are developed and tuned on 2016–2024 only and evaluated on the holdout once. The calib parameters (`calibLen`, `calibMinScored`, the margins) are set from dev-window evidence (D9a) before any holdout read; a change after a holdout read is a new trial. A holdout evaluation chart must start on the same first bar as the dev chart, so the degree path through the holdout is the one a live chart with that start would produce. (The Gate0 KD-02 canary still withholds bars on or after 2025-01-01 from its DVP stream; that is its own rule and is unchanged.)
- **DV-3.10.10 Where it cannot apply.** A stream that never reaches calibMinScored windows (a short chart, or a slow stream such as daily bars, where H = 60 bars is about 3 months and 60 windows is about 15 years) stays on d = 1 and warming for the whole run, and `DVP RUN` reports `warmEnd=NaN`. `Validate()` rejects `degMax = 1` with SelfCalibrate with a clear `ArgumentException` (DV-2.1) instead of running a calibrator with nothing to choose.

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4. Manifold, rails and nodes (`buildManifold`, `buildNodes`)

- **DV-4.1 Rows.** The span is z ∈ [−manSpan, +manSpan] in σ units of the current fit. The nominal row width is w_z = 2·manSpan/manRows. **Tick floor:** if w_z·σ < tick, then manRowsEff = max(1, ⌊2·manSpan·σ / tick⌋) and w_z = 2·manSpan/manRowsEff; otherwise manRowsEff = manRows. manRowsEff is journaled. Row r covers [−manSpan + r·w_z, −manSpan + (r+1)·w_z), centre z_r = −manSpan + (r + ½)·w_z.
- **DV-4.2 Volume deposit.** For each window bar i: lo_z = (L_i − ŷ(u_i))/σ, hi_z = (H_i − ŷ(u_i))/σ, using the **current** fit (the whole window is re-mapped on every refit). The bar's volume v_i is

spread over [lo_z, hi_z] in proportion to each row's overlap. The share outside ±manSpan goes to `tailVol`. A zero-range bar (hi_z = lo_z) deposits all of v_i into its row, or into tailVol if it falls outside. Rows are a fixed `double[manRows]` per stream, allocated in `Configure()` and cleared per refit (E5). `ValueAreaExpansion`, `PocIndex`, `Entropy` and `Prominence` are called on that array sliced to `manRowsEff`. Those APIs use `vols.Length`, so an exact-length array per `manRowsEff` value is kept in a small cache (allocated once per distinct value). §14.2-f still holds: `Accumulator.Add(..., mode 2)` is not used, because it spreads uniformly over tick-rounded keys, not by proportional overlap. With `splitFlow`, the same deposit is repeated with `RingBar.up` and `RingBar.down` into two more fixed arrays (diagnostic).

- **DV-4.3 Conservation.** Σ rows + tailVol = Σ v_i within 1e−9 relative (test DV-T9).
- **DV-4.4 POC and VA.** `_ProfileMath.ValueAreaExpansion(rows, vaPct / 100.0, out lo, out hi, out poc)` (the code multiplies the total by `percent`, so it takes a **fraction**; the catalogue's "70" is a percentage). VA = 70% of **in-span (manifold) volume**; tailVol is excluded, as in the book. The book's "70% value area grown outward from POC" matches `ValueAreaExpansion`'s two-row rule. POC ties use `PocIndex` (nearest to the volume-weighted mean row). pocZ = z_poc. vaHiZ = **outer** (upper) edge of row hi, vaLoZ = outer (lower) edge of row lo, consistent with the printed VA widths being whole multiples of the row height (Q9 resolved).
- **DV-4.5 Shape stats.**
- `entropy` = `_ProfileStats.Entropy(rows, EEntropyRows.TradedRows, 0).Hnorm`: H normalised by ln R over **occupied** rows, as in the book (B5 §2.1). There is no Miller–Madow correction, because the inputs are volume shares, not counts. Verified on main (`_ProfileStats` L193–213): `mmN = 0` skips the correction (`if (mmN > 0)`), so `Hmm`/`HnormMm` stay NaN. `Hnorm = H / ln(n)` with n = occupied rows for `TradedRows`, and NaN when n ≤ 1. `EntropyResult` is a struct, so no allocation. Unit case: rows {1,1,0,1,1} → **Hnorm** = 1.0 (raw H = ln 4 ≈ 1.386).
- `concentration` = rows[poc] / (Σrows / manRowsEff): the POC row over the mean row, printed as a multiple (book §1.3 / rev 3.1 §2.1). [GAP: the book does not print whether its mean runs over all span rows or only occupied rows. We use all `manRowsEff` rows (the "even spread" over the span) and journal the occupied-rows variant as `concentrationOcc`.]
- `inSampleCov2` = cov_in at 2σ (DV-3.6).
- **DV-4.6 Prominence adapter.** PR-D0b extracts the peak/trough prominence loop of `Gate0Profile.Nodes` (`_Gate0.Function.CS`, the part after the KDE) into `_ProfileMath.Prominence(double[] d, double prom, List<double[]> hvn, List<double[]> lvn)` inside `// <profile-math>`. It returns `{index, prominence}` pairs, and `Gate0Profile.Nodes` calls it. The change is behaviour-preserving: `tests/Gate0Tests` and `tests/ProfileTests` output must be unchanged. DVP calls it on the **raw** rows with prom = nodeProm. An HVN is a local maximum with topographic prominence ≥ nodeProm × max row. An LVN is the minimum between two adjacent HVNs with (lower peak − trough)/max ≥ nodeProm. `_VpHistogram` is not reused: it is bound to MultiCharts `IProfile` (§14.2-g). Smoothing is **off**: the book applies prominence 0.12 to the raw manifold rows (Q5 resolved). `Prominence` fills caller-owned lists; the DVP caller reuses preallocated `double[2]` node slots (DV-13.2).
- **DV-4.7 Node selection.** The POC row is removed from the HVN list, along with any HVN within 1 row of it. Per side of POC, keep the `maxNodes` HVNs and the `maxNodes` LVNs with the highest prominence (ties go to the nearer row to POC). Each node records z_node, prominence and side.
- **DV-4.8 Rails.** For bar t, a rail of residual level z has price **$P_t(z) = \hat{y}_{t-1}(u_t) + z \cdot \sigma_{t-1}$** (DV-3.1). Rail kinds and their z:

ELevelKind	z	family
`DvpPoc`	pocZ	manifold
`DvpVah` / `DvpVal`	vaHiZ / vaLoZ	manifold
`DvpHvn` / `DvpLvn`	z_node	manifold
`DvpSigma`	±k, k ∈ sigmaList	sigma
`DvpFib`	flat price (no z)	lattice
`DvpConf`	cluster price	(output of §6)

No z = 0 trend rail is published: the book's rail kinds are poc/vah/val/hvn/lvn/sigma (Q9 resolved). - **DV-4.9 Residual order.** The rails of one snapshot, sorted by z (lattice rails by their z-equivalent (P − ŷ(u_t))/σ), define "next rail in residual order" for targets: `NextRail(stream, fromZ, dir)`.

5. Lattice (`lattice`, Fib lattice builder)

- **DV-5.1 Pivots.** For each range j = 1..fibRanges, strength s_j = fibPivot·2^(j−1) (20, 40, 80). The pivots come from a `_PriceActionEngine` with `PriceActionConfig { enabled = true, dataStream = ds, layers = 1, strength = s_j, pivotMode = Fractal, minBreakoutLayer = 0, role = "DVP-S{s_j}" }` and the futures preset applied per field (`presetExtraPriceAction`). minBreakoutLayer = 0 is required because `PriceActionConfig.Validate` (L132) requires 0..layers−1, so layers = 1 fails with the default of 1; DVP never reads breakouts, so pivots are unaffected. Because `Canon()` includes minBreakoutLayer, DVP shares an ExtraPriceAction instance only when that host also uses 0; otherwise it gets its own instance (test T17SeparateInstance) (rev 5). Instances are found or shared by (dataStream, `Canon()`), as `findSharedPa` does. They are hosted by the DVP host code and stepped on their stream's closed bars, with `OnRoll` on `rollOn(ds)`: the same pattern as M-XMKT's hosted instances (`_CustomSignal` ~L1044 (PR #37 head)). The PriceAction library is not changed. Host order rules out ExtraPriceAction (`startExtraPa` runs before `startDvp`); XMKT `findSharedPa` pattern used (`findSharedPa(c, false)`, owned instances stepped in `stepDvp` unless hosted; rev 5, PR-D4). Book App E goes up to 4 ranges (20/40/80/160); fibRanges = 3 is the spec default, and 4 is allowed but not swept (Q10 resolved, §14.2-d).
- **DV-5.2 Anchor (MostRecentPivot).** For range j at bar t: B = the most recently **confirmed** pivot (confirmBar ≤ t−1) of either side, and A = the most recent confirmed pivot of the opposite side whose extremeBar is before B's. Both extremeBars must be within fibLookMult·s_j bars of t−1; if they are not, the range is empty. Range = |B − A|. Ties in B's confirmBar go to the later extremeBar, then to the high. A is the opposite-side confirmed pivot with the latest extremeBar before B's. The lookback is applied after selection and is inclusive (within ≤ fibLookMult·s_j bars); there is no fallback to an older pair, so an anchor outside it gives an empty range (rev 5).
- **DV-5.3 Levels.** `FIB_LEVELS = {0, .22, .5, .62, .71, .786, .886, 1, 1.414, 1.618, 2.618, 3.618, 4.236, −.146, −.272, −.414, −.764, −1, −2, −2.5, −2.62, −3, −3.5, −4}` (24 values, verbatim from rev 3.1 §2.2 and catalogue §2.4b; do **not** "correct" .22/.62/.71/−2.62 to textbook ratios). The level price is L(f) = B − f·(B − A): f = 0 is B, f = 1 is A, f > 1 extends past A, and f < 0 extends past B. [GAP: App E code unread; convention pre-registered, not book-verified. Q6 **DECIDED (Ettienne Hugo, 2026-10-23)**: pre-registered now; reading App E before PR-D4 is optional, and any change after that read is a new trial.]
- **DV-5.4 Merge and filter.** Within a range, greedily group levels in ascending price: a level joins the current group while it is strictly closer (<) than fibMerge·ATR_{t−1} to the group's first (lowest) member (no chaining) (rev 5). The merged price is the price of the member with the smallest |f − round(f)|, ties to the lower f. Drop levels further than fibNear·ATR_{t−1} from C_{t−1} (applied to each merged level's price after merging; |P − C_{t−1}| ≤ fibNear·ATR_{t−1} is kept, inclusive; rev 5). Levels from different ranges are never merged (each keeps its range id for nRanges).
- **DV-5.5 Flatness.** Lattice levels are flat prices. They are not re-evaluated on the curve.

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6. Confluence scorer

- **DV-6.1 Candidates.** All manifold rails (POC, VAH, VAL, HVN, LVN), all sigma rails, and all lattice levels of the snapshot, at their bar-t prices. The cone is excluded.
- **DV-6.2 Deterministic greedy clustering.** "Heaviest unclaimed" is not defined under a weightless count, so it is replaced by this deterministic, weightless order (Q11 resolved). For every unclaimed candidate c, let F(c) = the number of distinct families among unclaimed candidates within confTol·ATR_{t−1} of c, and K(c) = their count. The seed is the candidate with max F, then max K, then min |P − C_{t−1}|, then lowest price. The cluster is every unclaimed candidate within confTol·ATR of the seed. Mark them claimed and repeat until none are left. The cluster price is the seed price.
- **DV-6.3 Count.** `families` = the number of distinct families in the cluster (0..3). `nRanges` = the number of distinct lattice range ids. A cluster qualifies when families ≥ minFamilies = 3, which needs the lattice with ≥ 1 range. KD-09's "≥ 2 ranges" is applied **inside the KD-09 module**, never in the engine. Qualifying clusters are published as `DvpConf` LevelRefs with quality = families.
- **Rev 5.1 (S4):** Every rail occupies a candidate slot in publish order (a rail without a price is a non-candidate slot), so `quality` and journal families index rails 1:1.

- **DV-6.4 Point query.** `FamiliesAt(stream, price, tolAtr = confTol)` counts the distinct families and ranges among candidates within tolAtr·ATR of a given price, without clustering. RG-09/KD-09 use it on the host's event rail.
- **DV-6.5 Score.** `score` = `families` (0..3), the weightless family count of rev 3.1 §5.1. It is stored in `LevelRef.quality` for rails and confluence levels. A report-only sensitivity column `score077` is journaled next to it: the within-family maximum weight under VA edge 1.00 / POC 0.77 / every other kind 1.00, plus 1 for a sigma rail, plus 1 if nRanges ≥ 1. Neither value ever gates, ranks, sizes or selects. The published 1.00/0.80 weights are never computed. (Q12 moot.)

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7. Rolls, sessions and streams

- **DV-7.0 Roll detection (rev 5, instrument-profile amendment, M-INSTR IP-9.4). Waits on the DVP host PR (#36: `DvpConfig` host property, `startDvp`).** `rollOn()` returns false when `RollDetection == Off` (verified, `_CustomSignal` `rollOn`), and `IsRollWindow` is false when no ring exists. If `profile.rollMode == Detect` (futures) and `RollDetection == Off`, `Validate()`/`startDvp` logs one `ETraceLevel.Warning` "DVP: RollDetection Off — roll skip disabled" and **marks every stream diagnostic-only** (`DVP RUN` reports `rollDetect=Off`, skipShare = NaN); such a run cannot feed Holm. **ES with RollDetection Off is therefore unchanged from rev 4.** If `profile.rollMode == None` (stocks, ETFs, spot FX, crypto; no contract rolls by nature), no window can cross a roll: `crossesRoll` is always 0, **skipShare = 0** (not NaN), `diagnosticOnly` is decided by the other criteria only, `DVP RUN` reports `rollDetect=None`, and **the stream stays tradeable**. For ds ≥ 2, `rollOn(ds)` is valid only for the same root as Data1 (via `m_rollPending`). **A different root on Data2/3 is rejected in `Validate()`** (kept from rev 4). A Future Data2/3 under a non-future Data1 is rejected in `Validate()` (no detector for it). Under rollMode None, the RollDetection setting must be Off (a non-future with RollDetection on throws, DV-T27c), and `DVP RUN` then reports `rollDetect=None`, not Off. Consequential edits: DV-7.4 skipShare is 0 when rollMode None; DV-11.3 `rollDetect` ∈ {Series, Table, SeriesOrTable, Off, None}; DV-T27 unchanged; new DV-T27b (rollMode None: skipShare 0, not diagnostic-only, tradeable) and DV-T27c (non-future with RollDetection on throws, M-INSTR reason 13).
- **DV-7.1 crossesRoll.** A window crosses a roll if any bar i ∈ [1, n−1] of it has `RingBar.rollBar` (the first bar of a new contract). A window that starts on the roll bar is clean. On a crossing, the snapshot is **skipped**: no rails, `Rails()` returns empty, and the journal line has `crossesRoll=1`. Fits resume once n post-roll bars have accrued. `crossesRoll` and skip counting run on **every** closed bar from `RingBar.rollBar` (O(1) per bar, no fit), so skipShare is the same whether or not anyone queries (E10).
- **DV-7.2 Roll window.** Rail interactions whose trading day satisfies `IsRollWindow(day)` (roll session ±1) are journaled with `isRollWindow=1`, which is also set on the LevelRef. Consumers filter on it. (`IsRollWindow` is live only when a Phase-3 helper is enabled; DVP's `requireRing` call enables it.)
- **DV-7.3 ATR.** One `Gate0Atr(ATR_LEN)` per stream, fed on closed bars. `ResetPrev()` is called on `rollOn(ds)`. Snapshots use the value as of t−1.
- **DV-7.4 Skip share.** Per stream, skipShare = skipped / (eligible windows after warm-up), journaled in the end-of-run `DVP RUN` line. skipShare is 0 when the profile's rollMode is None (DV-7.0). **A stream with skipShare > 0.30 is diagnostic-only.** This is a run-level verdict used in analysis and reports. It cannot gate live bars, because it is only known at the end of the run. Arithmetic estimate only, not measured: if each roll invalidates n − 1 windows and a quarter has about 63 sessions, the expected shares are ≈ 4% (15m), ≈ 17% (60m), ≈ 63% (240m, about 6 bars/session, winLen 240) and ≈ 32% (240m, winLen 120). Measured values come from `DVP RUN` (DV-R3). Q4 **DECIDED (Ettienne Hugo, 2026-10-23)**: 240m is dropped from MT-07 and is diagnostic only.
- **DV-7.5 240m stream.** No winLen sweep (rev 4: the former {240, 120} sweep is removed). The 240m stream is **diagnostic only**; MT-07 trades 60m only. No winLen 90 trial (Q4 **DECIDED (Ettienne Hugo, 2026-10-23)**).
- **DV-7.6 ETH/RTH.** ETH is primary (all bars). RTH is the sensitivity arm, a second engine config built only from bars with `RingBar.rth`. `rthShare` = RTH volume / window volume is journaled and split into {< 50%, ≥ 50%}.
- **DV-7.7 Sessions, 24h charts and half-days.** Session boundaries are `tradingDayKey(t)` changes on the stream's own bars, never calendar dates. On the futures base this is the Globex date (verified: `_VolumeProfileFuturesBase` L356 overrides `tradingDayKey` with `GlobexDate`), so a 24h

chart's session starts at 18:00 ET. The RTH arm (DV-7.6) requires the futures base. On the generic base `isRthBar` always returns true (verified, `_CustomSignal` L516; futures override L512), so `rthShare` is journaled as NaN and the RTH arm is refused with a Warning. Half-day sessions (`todHalfDay`; base false at L546, futures override L491) are **not** removed from fit windows, because bars are bars. They are flagged `halfDay=1` on `DVP RAIL` and on A2-33 samples. The A2-33 once-per-session sample (Q8) is taken on the first closed bar of the new `tradingDayKey`, including half-days.

- **DV-7.8 KD-02 canary diagnostics (PR #68, report-only).** `Gate0CanaryDiag`, allocated in Canary mode only. No decision reads it.
- **DV-7.8.1 Counters.**
  - Every canary, one per RTH open, in this precedence: `devEnd` > `roll` (the roll session or the session after it) > `noPrior` > `noAtr` > `liveDays`. Also `rthDays` and `signals` (`enter()` calls).
  - KD-02, one bucket per DVP bar, in this order: `outsideRth`, `after1555`, `notLive`, `notTradeable`, `inPosition`, then the `Kd02Signal` reasons. Those are `noSnap`, `noRails`, `atrBad`, `insideBand`, and `pocNaN` / `railNaN` (`pocNaN` wins whenever the POC is NaN, including when the ±2σ band is NaN too), then `geomFail` and `tgtOnClose`; otherwise the bar is a signal.
  - Also `expired` (an unfilled signal cancelled at the next DVP bar; a later event, outside the sum) and `zeroVol1` = Data1 bars with volume ≤ 0 / Data1 bars fed.
  - Warnings, once per run: DVP window volume 0 (winLen consecutive zero-volume DVP bars), and a NaN POC or σ rail on a valid fit. Debug: one `G0-b KD02 decision` line per signal decision.
- **DV-7.8.2 G0-b canary line.** `G0-b canary(name) entries() signals() rollSkips() rollDays() sessions() rthDays() liveDays() skips{noPrior=,noAtr=,devEnd=,roll=[,dvpBars=,notLive=,outsideRth=,after1555=,notTradeable=,inPosition]} cumGross() cumNet()`. The bracketed fields appear for KD-02 only.
  - **Invariants** (tested in Gate0Tests and checkable from the log alone):
  - `dvpBars == outsideRth + after1555 + notLive + notTradeable + inPosition + noSnap + noRails + atrBad + insideBand + pocNaN + railNaN + geomFail + tgtOnClose + signals`
  - `rthDays == liveDays + noPrior + noAtr + devEnd + roll`
  - `rollSkips` counts sessions on or after DevEnd too, so `rollSkips ≈ 2 × rollDays` only approximately.

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8. Acceptance primitives

- **DV-8.1 Rail history.** Per stream, a ring of the last 8 snapshots' rail prices, so every bar is compared with **its own** t−1 rail. Trigger plus N further closes with N = 5 (sweep max) needs 6 snapshots and fits in 8 (t−1 … t−7). Assert N ≤ 6.
- **Rev 5.1 (S1, matching):** A rail is matched across snapshots by kind for POC/VAH/VAL, by z within 1e−6 for σ rails, and for HVN/LVN by the same-kind node nearest in z **provided |Δz| ≤ 0.25**. Otherwise the rail is absent in that snapshot (NaN), which breaks any run.
- **Rev 5.1 (S2, depth):** The ATR(t−1) history depth is max(refitEvery + 4, 10), so every rail-history snapshot (endAgo ≤ 8) carries its own ATR whatever the query order. Snapshots older than endAgo 9 are not part of the history contract.
- **DV-8.2 Trigger.** An up-trigger on rail R at bar t: C_t > P_t(R) + δ·ATR_{t−1} **and** C_{t−1} ≤ P_{t−1}(R) (no δ on t−1). A down-trigger mirrors it. δ = 0.10. **Rev 6:** on the first bar whose snapshot has a different degree from the t−1 snapshot (a monthly switch, DV-3.10.5), `Triggered` returns false, because the rail moved because the model changed, not because price did. A consumer that skips an entry for this logs it at Info (`skip degChange`). Acceptance runs already in progress continue, since each bar is compared with its own rail.
- **DV-8.3 Acceptance primitives.** The engine exposes:
- `RunBeyond(R, dir, deltaAtr)`: the number of consecutive closed bars, ending at t−1, each closing beyond its own rail + deltaAtr·ATR.
- `BackInside(R, dir)`: true if any close has come back inside by more than δ since the trigger.
- `Accepted(R, dir, N, countTrigger, deltaAtr)`: a thin wrapper over the two. **countTrigger = true:** the accepted run is the trigger close **plus N further closes**, each beyond its own rail + deltaAtr·ATR (N + 1 bars, trigger first). **countTrigger = false:** the run is N closes in total with the trigger as the first (trigger + N − 1 further closes).

The engine has no default convention. Each consumer passes its own catalogue text. PS-25 → `Accepted(R, dir, N = 2, countTrigger = true, deltaAtr = 0)` plus `!BackInside`: the trigger close plus 2 further closes beyond the rail, entry above the 2nd further close (Q1 **DECIDED (Ettienne Hugo, 2026-10-23)**). KD-06/KD-10 → countTrigger = false, N = 2, deltaAtr = δ on the confirmation bar ("C_t and C_{t+1} make N = 2", i.e. trigger + 1). - **DV-8.4** `closedThrough` (C beyond the rail with no δ), `heldBars` (RunBeyond with deltaAtr = 0), `distAtr` = (C − P)/ATR, and `verdict41` (held vs accepted on a symmetric 41-bar clock, PS-25 test 1) are reported for every interaction, next to MFE40/MAE40 in ATR (engine-side). - **Rev 5.1 (S3, verdict41 v0, [GAP]):** verdict41 is provisional, journal-only and never gates. For **trigger** events only, on the clock of the event bar + 40 bars: `accepted` if Accepted(N = 2, countTrigger = true, deltaAtr = 0) completes **on the event's own trigger** (2 bars after the event) before any close back inside its own rail by > 0.10 ATR; `failed` if such a close comes first; `held` otherwise. Accept events journal verdict41 = NaN. Journal field name: `verdict41v0`.

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9. HTF streams (MT-07, Probe P2)

- **DV-9.1** A stream ds > 1 in `streams` gets its own ring (`requireRing(ds, winLen + 8, "DVP")`), ATR, pivots and fit. It steps once per closed HTF bar (`BarsOfData(ds).Status == EBarState.Close`, `ringBarOf(ds)`).
- **DV-9.2** A Data1 bar reads the HTF snapshot of the last closed HTF bar via `_MtfMath.ClosedIndex` / `HtfClosedIndex(ds)`. Forming HTF bars are never used.
- **DV-9.3 Probe P2 gate.** The constant `DvpConst.P2_PASSED = false` (in `Canon()`). While it is false, HTF snapshots are computed and journaled but `Tradeable(ds)` returns false for ds > 1, and a Warning `DVP Data{ds} used before Probe P2` is logged once. It is flipped in a one-line PR once `_MtfProbe` P2 passes. **P2 passed 2026-11-02** (runs R1 `8356ba60`, R2 `05c8e4d9`, R3 `c003d107`, R4 `aa78186a`, `63e2d28d`; RHM on, @ES 1m + 60m; END stamps and per-series MaxBarsBack PASS; answers in missing-functionality-specs.md §3): `P2_PASSED = true` (PR-D10), so `Tradeable(ds)` for ds > 1 follows warming only and the Warning no longer fires.
- **Rev 5.2:** `Tradeable(ds)` never consults `diagnosticOnly`; that is a run-level verdict reported in DVP RUN only (DV-7.0, DV-7.4).
- **Rev 6:** `Tradeable(ds)` is also false while the stream is warming (DV-3.10.4), for every ds including Data1. Warming is known at bar time, so it can gate. A consumer logs each entry it skips because `Tradeable` is false at Info, with the reason (`warming`, `p2`).
- **DV-9.4 HTF acceptance.** `HtfAccepted(ds, R, dir, N ≥ 2)` applies DV-8.3 on HTF closed bars. `HtfNextRailDistAtr(ds, dir)` is the distance from C_{t−1} (Data1) to the next HTF rail in dir, in Data-ds ATR.
- **Rev 5.2:** The next rail is strictly beyond C(t−1) in dir (a rail exactly at the price is skipped). ATR is the Data-ds ATR from its t−1 snapshot. The result is NaN when dir = 0, there is no rail, or ATR ≤ 0 or NaN. `HtfAccepted` takes `countTrigger` and `deltaAtr` explicitly (no defaults, N ≥ 2).

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10. API (internal, on `_CustomSignal` / `_DvpEngine`)

```
internal _DvpEngine Dvp { get; }                          // null + one Warning when disabled
DvpSnapshot Snapshot(int ds)                              // as-of t−1 fit for bar t; null if no
fit / skipped
  .deg .n .sigma .sigmaAtr .r2 .yhatEnd .slopeEnd .slope1SigmaPerBar .endpointDrift
  .pocZ .vaLoZ .vaHiZ .entropy .concentration .tailVol .crossesRoll .lowR2
  .warming .degChanged                                     // rev 6, DV-3.10.4 / DV-3.8
double RailPrice(int ds, ELevelKind kind, double z)       // P_t(z); NaN if absent
IReadOnlyList<LevelRef> Rails(int ds)                     // Poc/Vah/Val/Hvn/Lvn/Sigma/Fib, tag
DV-10.2
IReadOnlyList<LevelRef> Confluence(int ds)                // DvpConf clusters, families==3
void FamiliesAt(int ds, double price, double tolAtr, out int families, out int nRanges)
LevelRef NextRail(int ds, double fromZ, int dir)          // residual order (DV-4.9)
double Z(int ds, double price)                            // (price − ŷ(u_t))/σ
bool Triggered(int ds, LevelRef r, int dir); int RunBeyond(int ds, LevelRef r, int dir, double
deltaAtr)
bool BackInside(int ds, LevelRef r, int dir); bool Accepted(int ds, LevelRef r, int dir, int N,
```

```
bool countTrigger, double deltaAtr)
bool HtfAccepted(int ds, LevelRef r, int dir, int N, bool countTrigger, double deltaAtr);
                                                    // N ≥ 2, no defaults (rev 5.2)
double HtfNextRailDistAtr(int ds, int dir)
bool Tradeable(int ds)                              // false if warming (rev 6); ds > 1: P2
int Degree(int ds)                                  // degree in effect at t−1 (rev 6)
void Cone(int ds, int h, out double lo, out double hi)  // [Diagnostic] never a level
```

- **DV-10.1** Each `LevelRef` uses `LevelRef.Line(kind, price, created = time of bar t−1 close, t0 = window first bar time, t1 = bar t−1 time, stream, tag, quality = families)`, with `crossesRoll` and `isRollWindow` set.
- **DV-10.2 Tag.** `"DVP{ds}:{t−1:yyyyMMddHHmm}:d{deg}:{kind}{z:+0.00}"` (lattice: `{kind}{f:+0.000}R{j}`).
- **DV-10.3 New ELevelKind values**, **appended** after `BootstrapPocZoneHi` so existing ordinals do not change: `DvpPoc, DvpVah, DvpVal, DvpHvn, DvpLvn, DvpSigma, DvpFib, DvpConf`. No switch over `ELevelKind` exists on main (checked), so appending is safe. `LevelRef.StudyType` (= kind.ToString()) gives the A2-33 type strings automatically.

10.4 What each consumer reads

Idea	Reads	Rule owned by the consumer (not the engine)
PS-25	`Rails(1)` (VAH/POC/HVN/LVN/σ), `Triggered`, `Accepted(N∈{2,3,5}, countTrigger = true)`, `Snapshot.sigmaAtr`, `NextRail`	entry on acceptance = trigger close + N further closes (countTrigger = true, deltaAtr = 0, `!BackInside`; Q1 **DECIDED (Ettienne Hugo, 2026-10-23)**); sigmaAtr ≤ 4; target = next rail; 18-cell sweep + 6-cell σ2/node block. **Primary cell CONFIRMED, DECIDED (Ettienne Hugo, 2026-10-23):** N = 2 (trigger + 2 further closes), event rail VAH/VAL, EX-05 trail + 41-bar time stop (catalogue PS-25, rev 3.1 ruling 6.1 item 2). It is frozen before any dev read; the other 17 cells are sensitivity cells.
MT-07	`HtfAccepted(ds = the 60m stream, N≥2)`, `HtfNextRailDistAtr`, `Tradeable(ds)`	HTF state as a filter for LTF entries. HTF = **60m only**; 240m is diagnostic, never traded (Q4 **DECIDED (Ettienne Hugo, 2026-10-23)**)
RG-09 / KD-09	`FamiliesAt(rail, 0.3)` after `Accepted`	≥ 2 ranges in-module; disagreement veto 0.3–1.0 ATR **[GAP]**; arm B families ≤ 1
RG-10	`Snapshot.slope1SigmaPerBar` (degree-1 shadow slope / σ₁), `Z(close)` history	gate if slope > 0.05 σ/bar; stand down winLen/4 bars after |z|>2 against the gate for ≥ 3 bars (off for KD-06)
KD-02 (canary)	`RailPrice(Sigma, +2)`, `RailPrice(Poc)`, `RailPrice(Sigma, +3)`	close > ŷ+2σ fade, target POC, stop 3σ + 0.1 ATR, time stops 19/40; **kill if net PF > 1.0**; `EGate0Canary.KD02` only
KD-06	`Accepted(Sigma ±2, N=2, countTrigger = false, deltaAtr = δ)` (trigger + 1)	EX-05 trail + 41-bar time stop
KD-10	node rails with |z_node| ≥ 2 after `Accepted(N = 2, countTrigger = false, deltaAtr = δ)`	stand-in for far nodes
A2-33	`Rails`/`Confluence` via the hook DV-12	G0-d types DvpPoc/Vah/Val/Hvn/Lvn/Sigma/Conf

Rev 6: every consumer checks `Tradeable(ds)` (false while warming, DV-3.10.4) and logs each skipped entry at Info with its reason (`warming`, `p2`, `degChange`). The engine never places or skips orders itself.

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11. Journal and trace

- **DV-11.1** **`DVP FIT`** (Journal, one per stream per refit when `journalFits`): `runId, configHash, stream, barTime, n, deg, bic[1..dCap], manRowsEff, rthShare, crossesRoll, sigma, sigmaAtr, r2, dw, skew, yhatEnd, slopeEnd, slope1, endpointDrift, pocZ, vaLoZ, vaHiZ, entropy, concentration, concentrationOcc, tailVol, inSampleCov2, lowR2, warming, degChanged,`

`coef{ȳ;ctr;scl;A0..Ad;B1..Bd;N0..Nd;C0..Cd}` (invariant culture, R round-trip format for coefficients). A skipped window writes `crossesRoll=1` and NaNs.

- **DV-11.2** **`DVP RAIL`** (Journal, when `journalRails`, on every trigger and acceptance): `railKind, z, price, families, nRanges, score, score077, distAtr, closedThrough, heldBars, accepted(N=2,3,5), verdict41, isRollWindow, halfDay`. Forward `MFE40 / MAE40` in ATR come from an engine-side 40-bar tracker. The interaction is also passed to `logSignal(tag, dir, price, stop = rail ∓ δ·ATR, context, taken: false)` with context keys `railKind, z, families, nRanges, score, deg, sigmaAtr`. (`logSignal` measures in R at `SignalHorizons` {5,15,30,60}. The DVP study strategy sets `SignalHorizons = {5,15,30,40,60}`, a strategy property, not a `DvpConfig` field; Q15 resolved.)
- **DV-11.3** **`DVP CREATED`** (once): streams, Canon(), degMode and the calib fields, the starting d per stream (1, warming, under SelfCalibrate), hosted PA labels, P2 state. **`DVP RUN`** (end of run): per stream: fits, skips, skipShare (NaN if rollMode Detect with RollDetection Off; 0 if rollMode None), `rollDetect` ∈ {Series, Table, SeriesOrTable, Off, None}, `diagnosticOnly` (skipShare > 0.30, or rollMode Detect with RollDetection Off), mean cov_in, mean cov_fwd, fallback count, and (rev 6) `calib{scored, rollDropped, badDropped, gridSkipped, rechecks, changes, warmEnd, dFinal}` (warmEnd = the bar time warm-up ended, NaN if never). In `Calibrate` the `DVP CAL` lines follow (D8). In `SelfCalibrate` with a journal sink the same `DVP CAL` lines are written from the run's scored windows, so one run gives both the online path and the offline pooled table (DV-T38, PR-D9a).
- **DV-11.4** Per-bar rows and lattice detail go to `Debug`.

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12. A2-33 harness hook

- **DV-12.1** With `LevelStudyConfig.enabled` and DVP enabled, the host calls `levelStudyAdd(r.StudyType, r.price, ageSessions: 1, quality: r.quality, key: r.tag)` for every `Rails(1)` and `Confluence(1)` level on the **first closed bar of each new `tradingDayKey`** (half-days included, flagged `halfDay`), using that bar's t−1 snapshot (the price is frozen at creation, the KD-11 convention). `_LevelStudy` assumes static levels aged in sessions, so a moving rail is sampled once per session (Q8 resolved; this avoids serial overlap). Types: DvpPoc, DvpVah, DvpVal, DvpHvn, DvpLvn, DvpSigma, DvpConf (DvpFib is reported but is not a G0-d type).
- **Rev 5.3 (S1):** `halfDay` is written on the per-session `DVP A2-33` Journal line, which joins the LevelStudy CSV by day (`tradingDayKey`, yyyy-MM-dd, one line per session), not by key; it is visible only at Journal trace level.
- **Rev 5.3 (S2):** Each level is registered after that bar's LevelStudy `Bar()`, so it is first evaluated on the next bar (never on its creation bar); the reference price is the creation-bar close.
- **DV-12.2** Levels with `crossesRoll` or `isRollWindow` are not added (A2-39c filter).

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13. Performance budget

- **DV-13.1** Disabled: one null check per `CalcBar`, no allocation, no rings, no PA instances (test DV-T22).
- **DV-13.2** Per refit per stream (n = 240, d ≤ 3, 64 rows, 3 ranges, ≈ 100 candidates): recurrence O(n·d), manifold O(n·rows spanned), VA/nodes O(rows), confluence O(K^2). Target **median ≤ 0.2 ms, p99 ≤ 1 ms** on the dev box, measured by `tests/Dvp` bench DV-T26 over 100k refits. **Zero allocations** per refit after warm-up (fixed row arrays and node slots, E5) (`GC.GetAllocatedBytesForCurrentThread` delta = 0).
- **DV-13.3** Memory per stream: ring winLen + 8 bars, 8 rail snapshots, work arrays ≈ 16 KB. Rev 6, SelfCalibrate: score ring ≈ 12 KB (DV-3.10.3), window copy winLen doubles (≈ 2 KB), ⌊projBars/calibStep⌋ + 2 pending coefficient sets (2 at the default calibStep) and the degree history (one byte per ring slot): ≈ 15 KB more per calibrator, all allocated in `Configure()`, zero allocations per bar. Hosted PA engines are the main cost (3 per stream). They are created only with `useFib`, and shared when their configs are equal.
- **DV-13.4** End to end, a 10-year @ES 15m backtest with DVP on Data1 and journal off may slow down by at most **1.00** (100%) versus DVP off, with `degMode = SelfCalibrate` (smoke test DV-R3). **DECIDED (Ettienne Hugo, 2026-11-01):** "Accept the slowdown and raise the timing budget" (rev 6.2; the rev 6 PROBE budget was ≤ 0.15). Evidence, PR-D9a host runs 4247ff31 (DVP on) / 5f06042a (DVP off), journal off, @ES 15m from 2016-11-03 20:30 ET through 2024 (189,478 chart bars): elapsedMs 6,683 on vs 3,775 off, slowdown = 6683 / 3775 − 1 = **0.77**. Those timing runs

have DVP on Data1 **and** Data2 (60m), so the measured slowdown is conservative against this Data1-only budget. (The journal-on run cb660d6a took 251,694 ms; the budget is stated with the journal off.)

- **DV-13.5 Self-calibration cost (rev 6, arithmetic estimate, PROBE).** Per grid sample (once per calibStep bars, default 60): copy 240 values, one nested fit to d = 3 (O(n·d), about 5–10 µs), and one pass for cov_in at every d (O(n·d), about 5 µs). Per bar: at most 2 pending entries (⌊H/calibStep⌋ + 2 with a smaller calibStep) × 3 degrees of a degree ≤ 3 evaluation and one square root (about 0.2 µs). Per recheck (monthly): 250 × 3 × 2 additions. Amortised, that is under 0.5 µs per bar, about 0.1 s over a 10-year @ES 15m run (≈ 230k bars), well under 1% of the DV-13.4 budget. When a cached snapshot fit exists for the grid window, the fit is reused (snapshot fits already nest to degMax), and only the cov_in pass is extra. Measured in DV-T26 (calibration on and off) and DV-R3.

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14. Open questions and conflicts

14.1 Decisions

Decided by Ettienne Hugo, 2026-10-23:

- **Q1 PS-25 acceptance counting. DECIDED (Ettienne Hugo, 2026-10-23).** "The trigger bar plus 2 more closes": acceptance = the trigger close plus 2 further closes beyond the rail; PS-25 passes `countTrigger = true, N = 2` (DV-8.3). KD-06/KD-10 keep their own text (trigger + 1, `countTrigger = false`).
- **PS-25 primary cell. CONFIRMED, DECIDED (Ettienne Hugo, 2026-10-23).** N = 2, event rail VAH/VAL, EX-05 trail + 41-bar time stop (§10.4).
- **Q4 240m stream. DECIDED (Ettienne Hugo, 2026-10-23).** Drop 240m: MT-07 runs on 60m only; 240m is journaled as diagnostic; no winLen 90 trial (DV-7.4, DV-7.5).
- **Q6 Fib direction. DECIDED (Ettienne Hugo, 2026-10-23).** Pre-register L(f) = B − f(B − A) now (DV-5.3); reading App E before PR-D4 is optional.

Decided by Ettienne Hugo, 2026-11-01 (5:58 PM ET):

- **Q16 Degree selection. DECIDED (Ettienne Hugo, 2026-11-01).** Replace the static per-symbol d* table (`DvpFrozenDegree` hashed in `Canon()`, `DevCoverageFrozen`, the offline freeze from a `_DvpEngineTest` Calibrate run) with **online self-calibration** while the strategy runs (DV-3.10). Reason: a static table does not carry over to forex, stocks or other futures. `degMode` = {SelfCalibrate (default), Fixed1, Calibrate}. Supersedes Q3.

Decided by Ettienne Hugo, 2026-11-01 (7:50 PM ET), with the rev 6 approval:

- **Q17 Parameter defaults. DECIDED (Ettienne Hugo, 2026-11-01).** Keep the PROBE defaults `calibLen` 250, `calibMinScored` 60, `calibMarginAbs` 0.01, `calibMarginRel` 0.02 for PR-D9. They are revisited only from the PR-D9a dev-window report, before any holdout read.
- **Q18 Score aggregation. DECIDED (Ettienne Hugo, 2026-11-01).** score(d) = |mean cov_fwd(d) − mean cov_in(d)| over the ring: the gap between the pooled means, the same as `Calibrate` and the book (not the mean of per-window gaps).
- **Q19 Slow streams. DECIDED (Ettienne Hugo, 2026-11-01).** Add the protected `DvpConfig` field `calibStep` (sample step in bars, default = projBars = 60, validated ≥ 1; §2, DV-2.1, DV-3.10.1), so 60m and daily streams can warm up. The horizon stays H = projBars.
- **Q20 Gating. DECIDED (Ettienne Hugo, 2026-11-01).** While warming, the stream is not `Tradeable` (DV-9.3), and a degree-change bar suppresses triggers (DV-8.2).
- **Q21 Chart start. DECIDED (Ettienne Hugo, 2026-11-01).** (Labelled Q6 in the approval message; relabelled Q21 in rev 6.1 so the rev 3 Q6 on Fib direction above keeps its label.) The standard chart start for dev and holdout evaluation runs is **2016-01-01**, because the degree path depends on the first bar the engine is fed (DV-3.10.8, DV-3.10.9).

Other remaining uncertainty is limited to the VERIFY and [GAP] items in the text.

Resolved (Analyst 1 recommendation adopted, rev 2):

Q	Decision	Where
Q2	Forward coverage uses the book's √(1+h) band, k = 2, H = 60; flat band report-only	DV-3.6
Q3	~~Frozen d* in a code constant table hashed in `Canon()`~~ **Superseded by Q16 (rev 6):** online self-calibration; `Canon()` hashes the parameters, not a table	DV-2.2, DV-3.10
Q5	Prominence on raw rows, no smoothing	DV-4.6
Q7	`Gate0Atr(20)`, Wilder	DV-1.10, DV-7.3
Q8	A2-33 sample once per session, keyed by `tradingDayKey`	DV-7.7, DV-12.1
Q9	Outer VA edges; no z = 0 rail; concentration = POC / mean row; `PocIndex` ties	DV-4.4, DV-4.5, DV-4.8
Q10	fibRanges = 3; 4 allowed, not swept	DV-5.1
Q11	Seed: most families, then most candidates, then nearest to C_{t−1}, then lowest price	DV-6.2
Q12	Moot: score is the family count; `score077` uses node 1.00	DV-6.5
Q13	RG-10 slope = degree-1 shadow slope over σ_1	§10.4
Q14	Journal counts as a consumer only when `engineSink(Journal) != null`; results path-independent	DV-0.1, DV-3.7/3.8
Q15	Horizon 40 in the study strategy's `SignalHorizons`, plus engine MFE40/MAE40 in ATR and `verdict41`	DV-8.4, DV-11.2

14.2 Conflicts found among the sources (resolution applied)

- a) **Analyst 2 review "Agree 5"** endorses weights 1.0/0.7/0.8. These are superseded by rev 3/3.1 (weightless count) and never used.
- b) **Analyst 2 P6/P7 entries** propose back-adjusted rolls, 5 new config fields and a KD-07 kill at PF > 1.1. Rev 3.1 overrules them: skip windows, no new fields, kill at PF > 1.0.
- c) **Analyst 1 P6/P7 review §8.8** says "N = 3 stays frozen". This is superseded by the N = 2 sign-off.
- d) **Book lattice of 4 ranges** vs fibRanges = 3. The spec keeps 3 and leaves 4 configurable (Q10).
- e) **Catalogue "ValueAreaExpansion(vols, 70, …)"**: the code expects the fraction 0.70 (DV-4.4).
- f) **Catalogue "Accumulator.Add (mode 2) on residual rows"**: mode 2 is uniform over tick keys, not proportional overlap. The spec uses `AddRow` (DV-4.2).
- g) **Catalogue "_VpHistogram node logic"**: that class is `IProfile`-bound. The reusable prominence code is in `Gate0Profile.Nodes` (KDE-based), and it is extracted in PR-D0b.
- h) **Catalogue "_RangeProfile ring pattern"**: the Phase-3 `requireRing`/`_BarRing` is the current mechanism and is the one used.
- i) **PS-25 vs KD common acceptance wording**: resolved by Q1 (DECIDED (Ettienne Hugo, 2026-10-23)); PS-25 = trigger + N further (countTrigger = true), KD-06/KD-10 = N including the trigger (countTrigger = false).
- j) **240m sparsity** makes the 30% rule bite: resolved by Q4 (DECIDED (Ettienne Hugo, 2026-10-23)), 240m diagnostic only.
- k) **Line numbers** cited in the Analyst 1 review (`IsRollWindow` ~L1100, VPF-4 ~L1391–1417) still match main (`IsRollWindow` L1100, `_RollDetector` L1398). **Confirmed.** The catalogue's `HtfClosed`/`HtfClosedBar` exist (L1330–1334). **Confirmed.** (The `PriceActionConfig`/`m_pa` block is at L383–397, not L379–397.)
- l) **Rev 1's report score vs rev 3.1** (E1): replaced by the family count.
- m) **Rev 1's flat-band forward coverage vs the book's leveraged definition** (E6): the leveraged definition is adopted.
- n) **KD-06 vs PS-25 δ placement** (E4): δ goes on the trigger only (PS-25) or on the confirmation bar (KD-06/KD-10), passed per consumer.
- o) **Rev 3.1 "no new fields" vs rev 6's four `calib*` fields**: resolved by Q16 (DECIDED (Ettienne Hugo, 2026-11-01)). The fields are protected `DvpConfig` settings with defaults, not `[Input]`s, and they are hashed in `Canon()`. The "no new fields" ruling still holds for everything else (b).
- p) **`_TestGate0` header and `Gate0.Kd02Config` summary on main** describe the old freeze plan ("d* FREEZE … goes into DvpFrozenDegree in PR-D9"). They are superseded by rev 6 and are corrected in a doc follow-up (§16 item 14). No code is edited by this spec revision.

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15. Tests (`tests/Dvp`, ProfileTests style)

`gen.sh` extracts `// <dvp-core>`, `// <profile-math>`, `// <level-ref>` and `// <dvp-pa>` (`DvpPaBridge`). `_PriceActionEngine.Function.CS` (with `PriceActionTests/McShim.cs`) is linked whole in the tests/Dvp csproj (rev 5). `Program.cs` uses `Check(bool, string)` with pass/fail counts. `run.sh` runs `dotnet run -c Release` with LangVersion 7.3. `reference.py` (numpy `lstsq` / QR) generates the fixture CSVs under `tests/Dvp/fixtures/`, following the RegimeTests reference pattern.

- **DV-T1 Recurrence vs reference.** 50 random series (n ∈ {12, 30, 240, 1000}, d = 0..3, w = 1 and w = v): ŷ, σ, SSE and h match `reference.py` to 1e−9 relative.
- **DV-T2 Shift/scale invariance.** y + 10^6 gives the same residuals and σ to 1e−9; y × 4 scales σ by 4.
- **DV-T3** SSE is non-increasing in d. SSE ≥ 0. Forcing N_k ≤ 0 triggers the fallback path and journals it.
- **DV-T4 Exact recovery.** A cubic y with no noise gives SSE ≈ 0 and exact ŷ(u_t). A linear y with d = 3 gives $C_2 = C_3 \approx 0$.
- **DV-T5 Coverage drill.** Trend plus Gaussian noise, 2,000 windows: mean cov_in(2σ) ∈ [0.94, 0.965].
- **DV-T6 Leverage.** Σ_i h(u_i) = d + 1 (w = 1). h is minimal near u = 0 and h(u_t) > h(1).
- **DV-T7 BIC** values equal the formula. BIC never changes the chosen d.
- **DV-T8 Degree.** dCap for n = 12, 16, 24, 240. Fixed1 gives d = 1. Calibrate fits every d and reports d = 1. SelfCalibrate reports the calibrator's degree, always ≤ dCap. n < 12 gives no fit. (Rev 6: the DevCoverageFrozen and table-lookup cases are removed.)
- **DV-T9a Shape stats.** Entropy rows {1,1,0,1,1} → Hnorm = 1.0 with `mmN = 0` (HnormMm NaN). Concentration on a known histogram equals rows[poc]/(Σ/manRowsEff); `concentration0cc` is journaled.
- **DV-T9 Manifold deposit.** A bar spanning 2.5 rows splits 0.2/0.4/0.4 as its overlap dictates. Zero range goes to one row. An outside bar goes to tailVol. Conservation holds (DV-4.3).
- **DV-T10 Tick floor.** σ = 0.5 tick gives manRowsEff = max(1, ⌊6·0.5⌋) = 3, and it is journaled.
- **DV-T11 VA.** Equals `ValueAreaExpansion(rows, 0.70)`. Passing 70 is rejected by Validate/asserted. POC tie rule.
- **DV-T12 Prominence.** (a) `tests/Gate0Tests` and `tests/ProfileTests` outputs are bit-identical before and after PR-D0b. (b) Synthetic bimodal rows give 2 HVN and 1 LVN; at prominence 0.11 vs 0.13 the threshold is respected. (c) maxNodes per side and POC exclusion.
- **DV-T13 Rails.** P_t(z) uses the t−1 fit at u_t = 1 + 2/(n−1). Rail kinds and tags match DV-10.2.
- **DV-T14 Truncation audit.** Snapshots at bar t computed on a series truncated at t−1 are bit-identical to the full-series run, for 500 random t.
- **DV-T15 Shuffled future.** Permuting bars > t−1 leaves every bar-t output unchanged.
- **DV-T16 Rolls.** A synthetic splice: windows containing it are skipped. A window starting on the roll bar is clean. Fits resume after n post-roll bars. skipShare and `diagnosticOnly` at 0.30 ± one window. ATR `ResetPrev`. `rollOn(1)` sets `rollBar` only on the closed first bar of the new contract.
- **DV-T16b Path independence.** Journal-on and journal-off runs (and runs with random query patterns) give identical snapshots, endpointDrift and skipShare on 100 random bars, including `refitEvery > 1`.
- **DV-T17 Lattice.** A pivot is unusable before `confirmBar`. Anchors outside fibLookMult·s are dropped. Merge at 0.40 ATR keeps the lowest-|f−round f| member. The ±14 ATR filter applies. Ranges never merge across.
- **DV-T18 Confluence.** (a) Manifold + sigma only gives 2 (no qualify); adding a lattice line gives 3 (qualifies). (b) Two lattice lines from one range give nRanges = 1; from two ranges, 2. (c) Changing the report weights never changes families or the cluster set, and a source scan finds no 0.80 / 0.7 weight constant in `// <dvp-core>`. (d) The seed order is deterministic under input permutation. (e) The tolerance boundary at 0.30 ATR is inclusive.
- **DV-T19 Acceptance.** N = 2/3/5 with a moving rail: a bar that beats the old rail but not its own rail does not count. The trigger needs C_t > rail + δ and C_{t−1} ≤ rail (no δ). Both counting readings and both deltaAtr values; `BackInside`; assert N ≤ 6.
- **DV-T19b PS-25 counting (Q1 DECIDED).** With `countTrigger = true, N = 2`: trigger + 1 further close beyond → not accepted; trigger + 2 further closes → accepted on the 2nd further close; a 2nd further close beyond the old rail but not its own → not accepted. With `countTrigger = false, N`

`= 2` (KD-06): trigger + 1 further close → accepted. N = 5 with countTrigger = true uses 6 snapshots.

- **DV-T20 Cone.** Matches the formula. It never appears in `Rails`, `Confluence` or `FamiliesAt`.
- **DV-T21 HTF.** A forming Data2 bar never changes a Data1 snapshot (truncate the HTF series at the forming bar: bit-identical). A same-root Data2 roll is detected; a different root is rejected by Validate. The Data1 read maps to the last closed HTF bar. `Tradeable(2)` is false while P2_PASSED is false, with one Warning.
- **DV-T22 Laziness.** Disabled: `Dvp == null`, one Warning on query, no ring, no PA. Enabled with no query and journal off: the snapshot fit counter is 0. Under SelfCalibrate the calibration fit counter equals the number of sampled grid windows, and no manifold, rail or lattice is built (rev 6, DV-0.1).
- **DV-T23 Config.** Validate ranges, including the rev 6 calib fields, `calibMinScored > calibLen`, and SelfCalibrate with degMax = 1 (each throws `ArgumentException` with a clear message). `Canon()` is stable and includes the constants, `CALIB_K`, `CALIB_RECHECK` and the four calib fields, with no degree table. Changing any calib field changes `Canon()`.
- **DV-T24 RG-10 primitives.** The slope sign and units (σ_1 per bar) on a synthetic ramp.
- **DV-T25 Recalc.** Running twice gives identical journals.
- **DV-T26 Perf.** The DV-13.2 budget and zero allocations (no `Accumulator.Dense()` on the path).
- **DV-T27 RollDetection Off (rollMode Detect).** One Warning, every stream diagnostic-only, skipShare reported as NaN (not 0).
- **DV-T27b rollMode None** (stock/ETF/FX/crypto profile, waits on #36): no Warning, skipShare = 0, stream not diagnostic-only, tradeable.
- **DV-T27c** Non-future with RollDetection on: throws (M-INSTR reason 13).
- **DV-T28 24h session.** A synthetic Globex session crossing midnight is one session (one A2-33 sample, one session boundary).
- **DV-T29 Half-day.** A half-day session gives one A2-33 sample with `halfDay=1`, and its bars stay in fit windows.
- **DV-T30 Self-calibration, no look-ahead (rev 6).** (a) A window sampled at s is scored only on bar s + H, and the scorer reads no bar > t on bar t (instrumented source). (b) Truncation: the degree path up to bar t on a series truncated at t is identical to the full-series path, for 200 random t. (c) Permuting bars > t leaves the degree in effect at t unchanged. (d) An entry whose horizon is incomplete at the end of the data is never scored.
- **DV-T31 Warm-up.** Before calibMinScored scored windows: d = 1, `warming` true, `Tradeable(ds)` false, and exactly one `DVP DEGREE warming` Warning per calibrator. Reaching calibMinScored mid-month changes nothing until the next recheck. A series too short to warm up stays on d = 1 with `warmEnd=NaN` in `DVP RUN`.
- **DV-T32 Monthly-only switching.** A synthetic series whose best degree flips mid-month: d changes only on the first closed bar of a new `tradingDayKey` month, never between rechecks. A Globex Sunday-evening session that opens a new month is the recheck bar (futures base). A month with no bars rechecks on the next bar. The stream's first bar is never a recheck.
- **DV-T33 Hysteresis.** A score gap just below max(calibMarginAbs, calibMarginRel × score(c)) does not switch; just above, it switches. Separate cases where the absolute margin binds and where the relative margin binds.
- **DV-T34 Ties.** Exact equal scores: the candidate is the lower d. Leaving warm-up with tied scores keeps d = 1. An incumbent d = 3 tied by d = 2 stays at 3 (the margin is not met).
- **DV-T35 Roll-crossing horizons.** A splice inside a pending horizon drops that window whole (`rollDropped` + 1, not in the ring). A grid position whose window crosses a roll is not sampled (`gridSkipped` + 1), and the grid does not shift. Under rollMode None nothing is dropped.
- **DV-T36 Recalc reproducibility.** Two runs give byte-identical `DVP DEGREE` lines and degree paths. Journal-on, journal-off, random query patterns and refitEvery ∈ {1, 4} give identical degree paths, snapshots and endpointDrift (E10). After a switch, endpointDrift equals the value computed with both fits at the new degree, and `degChanged=1` appears once.
- **DV-T37 Journal formats.** `DVP DEGREE sel stream() winLen() from() to() scores[...] n()` at Journal on every recheck, with dCap scores in invariant `R` format (`NaN` when empty). `DVP DEGREE change ...` at Info only on a change. The warming Warning text. A null sink formats no string (allocation check). One line per `File.AppendAllText`.
- **DV-T38 Online vs offline parity.** With calibLen at least the number of scored windows, the ring means equal the `DVP CAL` pooled means of the same run to 1e−12, and the `DVP CAL` table from a

SelfCalibrate run equals the one from a Calibrate run over the same bars before `HOLDOUT_FROM`. The degree-change trigger suppression of DV-8.2 is tested here too.
- **DV-T39 Trading-day year buckets.** A window ending on the evening of Dec 31 (Globex trading day in the next year) is counted in the next year's `DVP CAL` bucket; a window ending on the afternoon of Dec 31 stays in the old year. `Calibrate`'s `HOLDOUT_FROM` test still uses the bar time.

15.1 Regression fixtures

- **DV-R1 Golden journal.** A synthetic 15m series of 5,000 bars with two splices (fixture CSV). `DVP FIT/RAIL` lines are byte-identical to `fixtures/golden_fit.txt`. Rev 6: regenerated in PR-D9 (new `Canon()` hash, `warming`/`degChanged` fields, `DVP DEGREE` lines). The fixture is long enough to leave warm-up and contain at least one recheck.
- **DV-R2 Reference parity.** `reference.py` reproduces DV-R1's σ, pocZ, vaLoZ/HiZ and the node set to 1e−9 / exact.
- **DV-R3 Host smoke test (PROBE).** `_DvpEngineTest.Strategy.CS` on @ES 15m (+ Data2 60m), 2 years, journal on. Check skipShare per stream, mean cov_in ≈ 0.95, endpointDrift distribution, and the slowdown budget (≤ 1.00, DV-13.4, DECIDED rev 6.2, with SelfCalibrate on). Rev 6: the same host also runs the self-calibration validation of PR-D9a (report-only).
- **DV-R4 Canary.** `EGate0Canary.KD02` in G0-b must have net PF ≤ 1.0 (> 1.0 = bug).

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16. PR plan (each PR small and independently reviewable; merged by Ettienne)

1. **PR-D0a: `ELevelKind` DVP kinds.** Append the 8 kinds (DV-10.3). No behaviour change. Build + all test suites.
2. **PR-D0b: Prominence extraction.** `_ProfileMath.Prominence` with `Gate0Profile.Nodes` delegating to it. DV-T12a (bit-identical Gate0/Profile tests).
3. **PR-D1: Fit core.** `_DvpEngine.Function.CS` `// <dvp-core>`: `DvpConfig` (Validate/Canon), `DvpFit` (recurrence, σ, SSE fallback, leverage, BIC, degree modes, shadow d=1, cone), `DvpFrozenDegree` (empty; removed in PR-D9, rev 6). `tests/Dvp` scaffold + `reference.py`. DV-T1–T8, T20, T23. Not wired.
4. **PR-D2: Manifold and rails.** Fixed row array (E5), tick floor, VA, entropy/concentration (E2/E3) with unit cases, nodes, rails, LevelRef/tags. DV-T9a, T9–T13.
5. **PR-D3: Host wiring, Data1.** `protected DvpConfig`, startDvp/stepDvp in `startEngines`/`stepEngines`, `requireRing`, ATR, roll skip, lazy cache with path independence (E10), RollDetection-Off warning and diagnostic flag (E8), tradingDayKey/half-day handling (E9), `DVP FIT`/`CREATED`/`RUN` journal. DV-T14–T16, T16b, T22, T25, T27–T29, DV-R1, and the `rollOn(1)` rollBar test.
6. **PR-D4: Lattice.** Hosted PA pivots (DV-5.1), anchors, levels, merge/filter. DV-T17.
7. **PR-D5: Confluence + acceptance.** Clustering, `FamiliesAt`, report score, trigger, `RunBeyond`/`BackInside`/`Accepted(countTrigger, deltaAtr)` (E4) with tests for both counting readings (DV-T19, DV-T19b), rail history, `DVP RAIL` + `logSignal` + MFE40/MAE40, RG-10 primitives. DV-T18, T19, T24.
8. **PR-D6: HTF streams.** Data2/Data3, P2 gate, HtfAccepted/HtfNextRailDistAtr, no winLen sweep (240m diagnostic only, Q4). DV-T21, including the forming-HTF-bar truncation test and the same-root Data2 roll test.
9. **PR-D7: A2-33 hook.** DV-12. A LevelStudyTests case with DVP type strings.
10. **PR-D8: Canary + calibration.** Stays before PR-D9: the canary must run on the engine before any consumer. `EGate0Canary.KD02` **appended** after `PS03` (Gate0 calls `// <dvp-core>`), `degMode = Calibrate` coverage journal for KD-08 (`DVP CAL`; diagnostic only since rev 6, it no longer sets a frozen d*). DV-R3 smoke and DV-R4.
11. **PR-D9: Online degree self-calibration (rev 6, DV-3.10).** `EDvpDegMode { SelfCalibrate, Fixed1, Calibrate }`: SelfCalibrate replaces DevCoverageFrozen at ordinal 0 and is the default. Add the four `calib*` fields (§2) with Validate rules (DV-2.1). Remove `DvpFrozenDegree` and its `Canon()` entry, and hash `CALIB_K`, `CALIB_RECHECK` and the calib fields instead. The calibrator in `// <dvp-core>`: fixed sample grid, pending queue, rolling score ring, monthly recheck on `tradingDayKey`, hysteresis, warm-up (d = 1, `Tradeable` false, one Warning), and degree history for lazy recomputes. `DvpCalib` (D8) moves to the same fixed grid and shares the scoring code. endpointDrift at the current degree, plus `degChanged` (DV-3.8). Degree-change trigger suppression (DV-8.2). `DVP DEGREE`

lines and `DVP RUN calib{...}` (DV-11.3). Gate0's `Kd02Config` inherits SelfCalibrate, and warming DVP bars count as `notTradeable` (DV-7.8.1). Tests: DV-T8, T22, T23 rewritten; DV-T30–T38 new; DV-R1 regenerated; HostTests updated for the removed mode. This changes `Canon()`, so every earlier journal hash changes; the PR body must say so. No `[Input]`. After this, consumer sandboxes (PS-25, MT-07, RG-09/10, KD-06/10) may start.

12. **PR-D9a: `_DvpEngineTest` host (Coder, `feat/dvp-d9a-engine-test`).** Kept as the DV-R3 host smoke test (@ES 15m + Data2 60m, journal on; skipShare, cov_in, endpointDrift, slowdown ≤ 1.00 per DV-13.4 rev 6.2). Its `dvpFeed`/`dvpFinish` hooks on `_CustomSignal` add no `DvpConfig` field, so `Canon()` is unchanged. Rev 6 adds a **self-calibration validation, report-only:** @ES 15m, chart start 2016-01-01 (Q21 DECIDED (Ettienne Hugo, 2026-11-01)), 2016–2024 (`dvpFeed` withholds every bar on or after 2025-01-01), `degMode = SelfCalibrate`, journal on. Compare the online degree path (the `DVP DEGREE sel` lines) with the offline `Calibrate` result over the same windows (the `DVP CAL` pooled d* and per-year d*, which DV-T38 shows equal a separate Calibrate run). Report: share of months where the online d equals the offline pooled d*, number of switches, warm-up end, the distribution of the score gaps at rechecks, and whether the PROBE defaults look right (Q17). No table, no gate, no `Canon()` change. Dependencies: the smoke part needs D8 only; the validation needs D9 (rebase on D9, or add the comparison as a follow-up commit once D9 merges).
13. **PR-D10 (when P2 passes): flip `P2_PASSED`.** One line plus the probe evidence. Unchanged by rev 6. It does not depend on D9 or D9a. **Done 2026-11-02** (rev 6.3).
14. **Doc follow-up (rev 6, no code change).** Bring the comments and docs that describe the old freeze plan in line with DV-3.10: the `_TestGate0.Strategy.CS` header (KD-02 "d* is not frozen yet (PR-D9)" and "d* FREEZE (Ettienne Hugo, 2026-10-31) ... goes into DvpFrozenDegree in PR-D9"), the `Gate0.Kd02Config` summary in `_Gate0.Function.CS`, the `_DvpEngine.Function.CS` header and `DvpConst.HoldoutFrom` comments (where D9 does not already touch them), and engines-manual §5.31 (DevCoverageFrozen / `DvpFrozenDegree.Table` / "DVP DEGREE not frozen"). It can go in with D9 or straight after it.